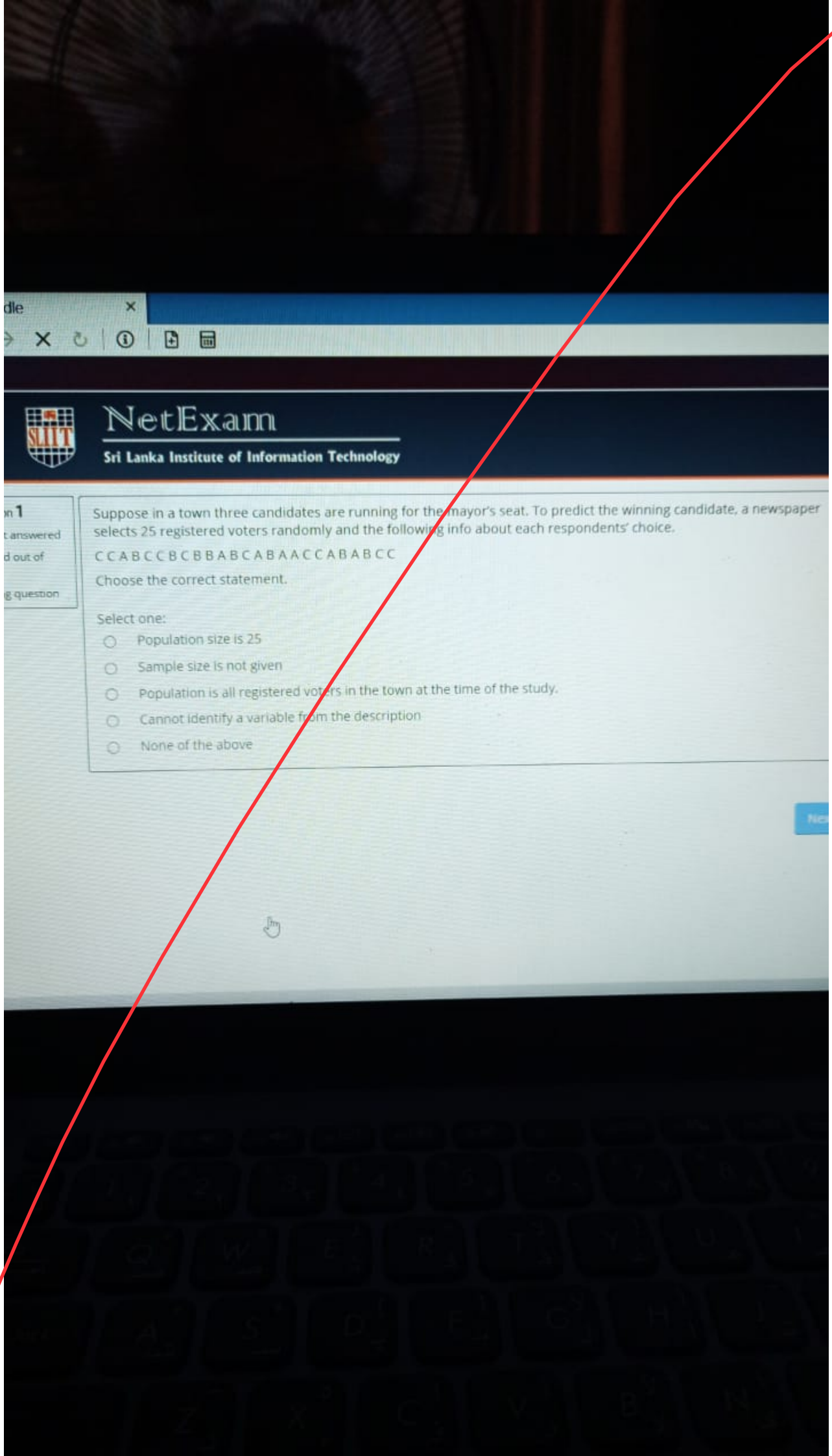
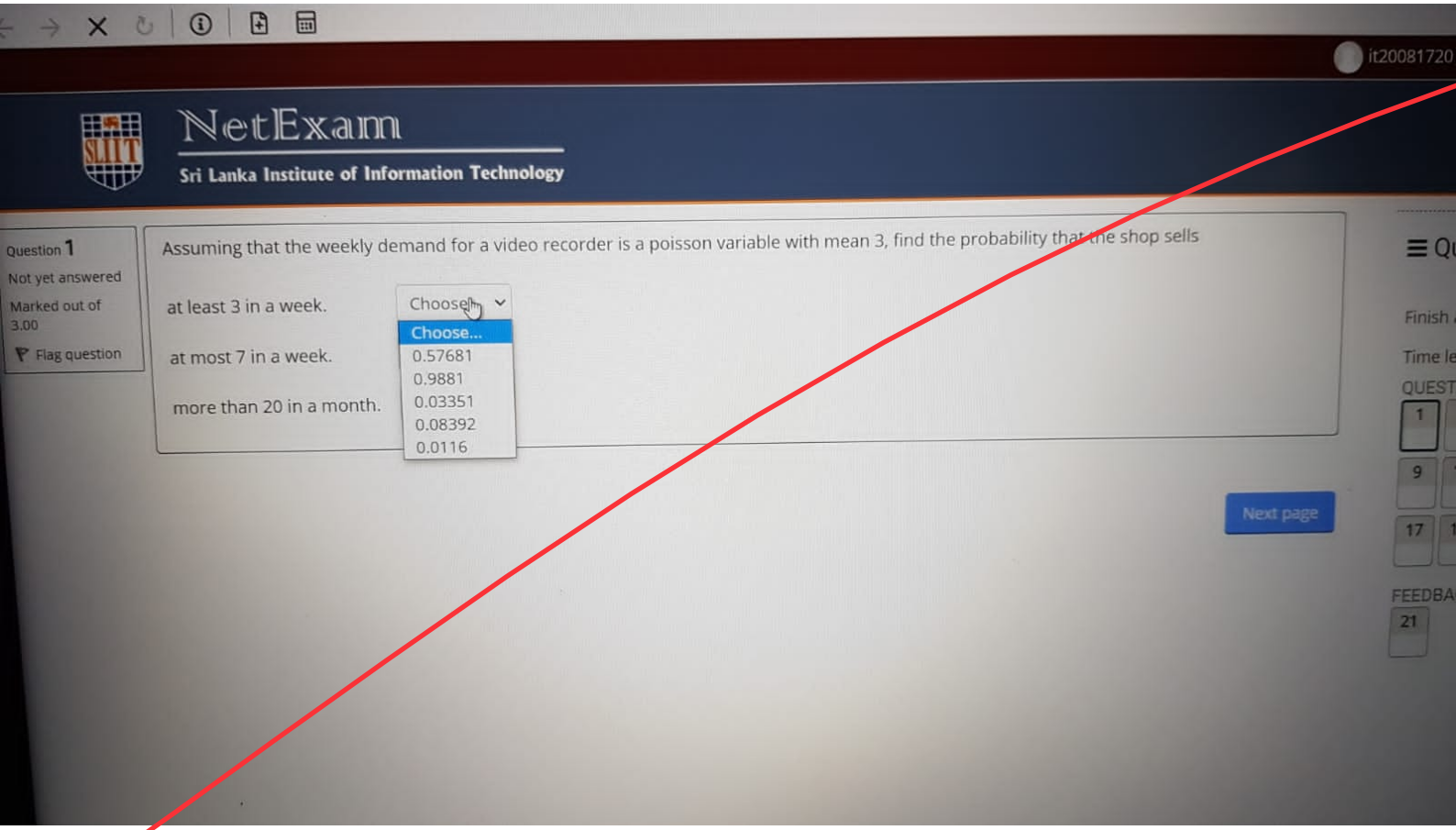

NetExam
Sri Lanka Institute of Information Technology
Suppose in a town three candidates are running for the mayor's seat. To predict the winning candidate, a newspaper selects 25 registered voters randomly and the following info about each respondents' choice.
C C A B C C B C B B A B C A B A A C C A B A B C C
Choose the correct statement.
Select one:
Population size is 25
Sample size is not given
Population is all registered voters in the town at the time of the study.
Cannot identify a variable from the description
None of the above


it20081720
NetExam
Sri Lanka Institute of Information Technology
Question 1
Not yet answered
Marked out of 3.00
Flag question
Assuming that the weekly demand for a video recorder is a poisson variable with mean 3, find the probability that the shop sells
at least 3 in a week.
Choose...
at most 7 in a week.
more than 20 in a month.
Choose...
0.57681
0.9881
0.03351
0.08392
0.0116
Next page
Finish
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1
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17
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21

Sri Lanka Institute of Information Technology

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Flag question

The number of industrial injuries per working week in a particular factory is known to follow a Poisson distribution with mean 0.5. Find the probability that,

in particular week there will be less than 2 accidents. Choose...

in particular week there will be more than 5 accidents. Choose...

in a 3 week period there will be no accidents.

Choose...
0.54476
0.9098
0.77687
0.22313
0.00001

Next page

NetExam

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E(X) and $E(X^2)$ of a discrete random variable are -1.4 & 2 respectively. What is V[X]?

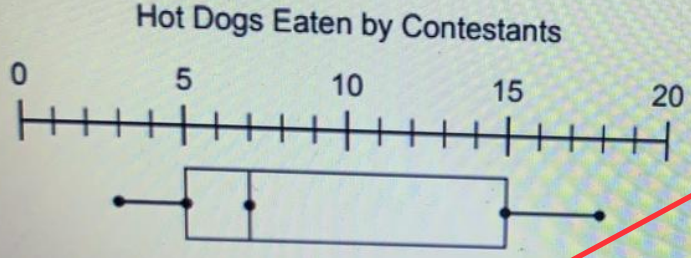
Select one:

- -0.4
- 0.4
- 1.4
- 0.04
- None of the above

Sri Lanka Institute of Information Technology

1
nswered
out of
question

According to the box-and-whisker plot, what is the maximum number of hot dogs eaten in the hot dog eating contest?



Select one:

- 15
- 3
- 7
- 18
- 20

NetExam

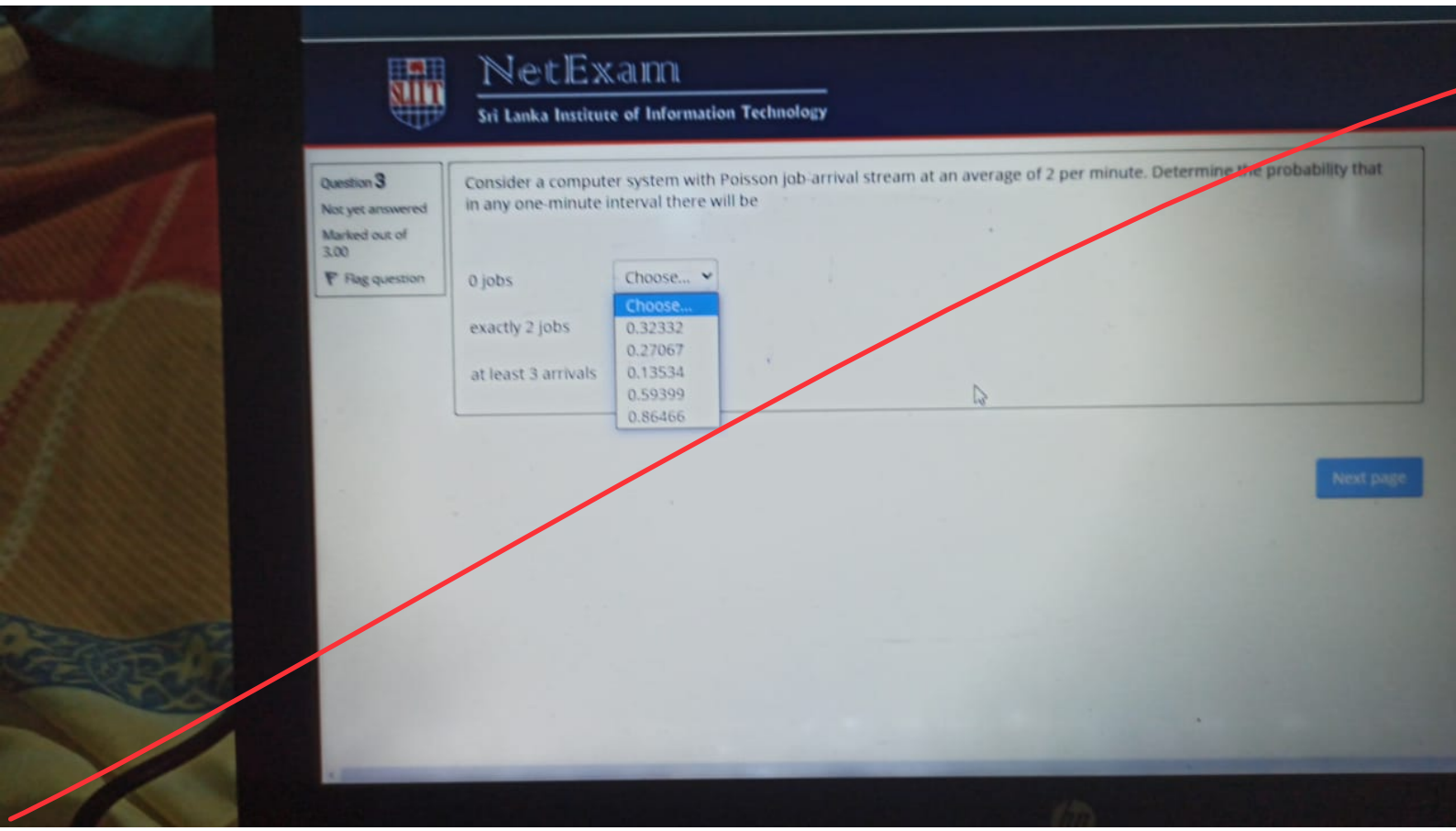
Sri Lanka Institute of Information Technology

The statistic NOT required for a box plot is.

Select one:

- Mean
- 1st quartile
- Median
- 3rd quartile
- None of the above

Next p


NetExam
Sri Lanka Institute of Information Technology
Question 3
Not yet answered
Marked out of 3.00
Flag question
Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be
0 jobs
exactly 2 jobs
at least 3 arrivals
Choose...
Choose...
0.32332
0.27067
0.13534
0.59399
0.86466
Next page

NetExam

Sri Lanka Institute of Information Technology

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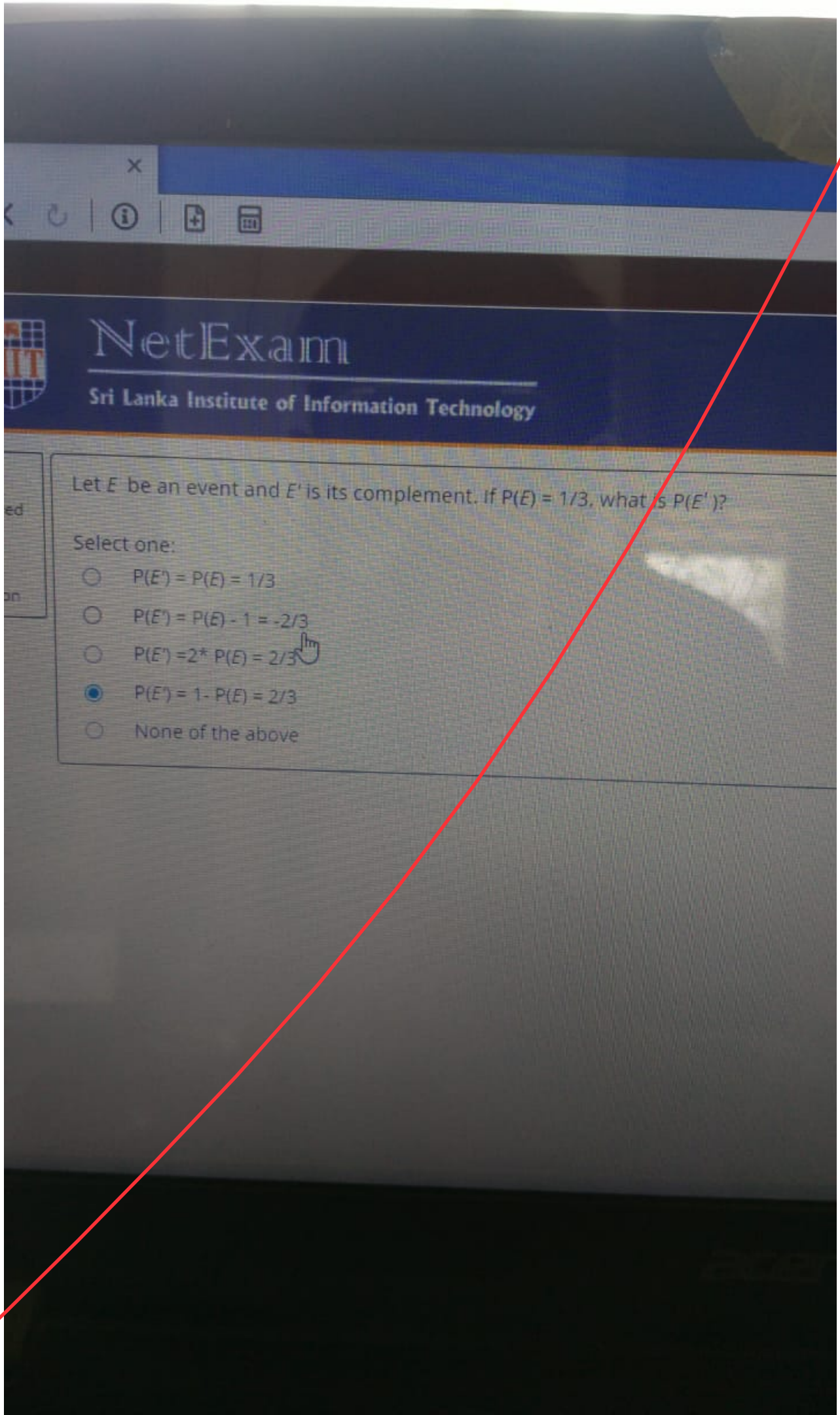
estion

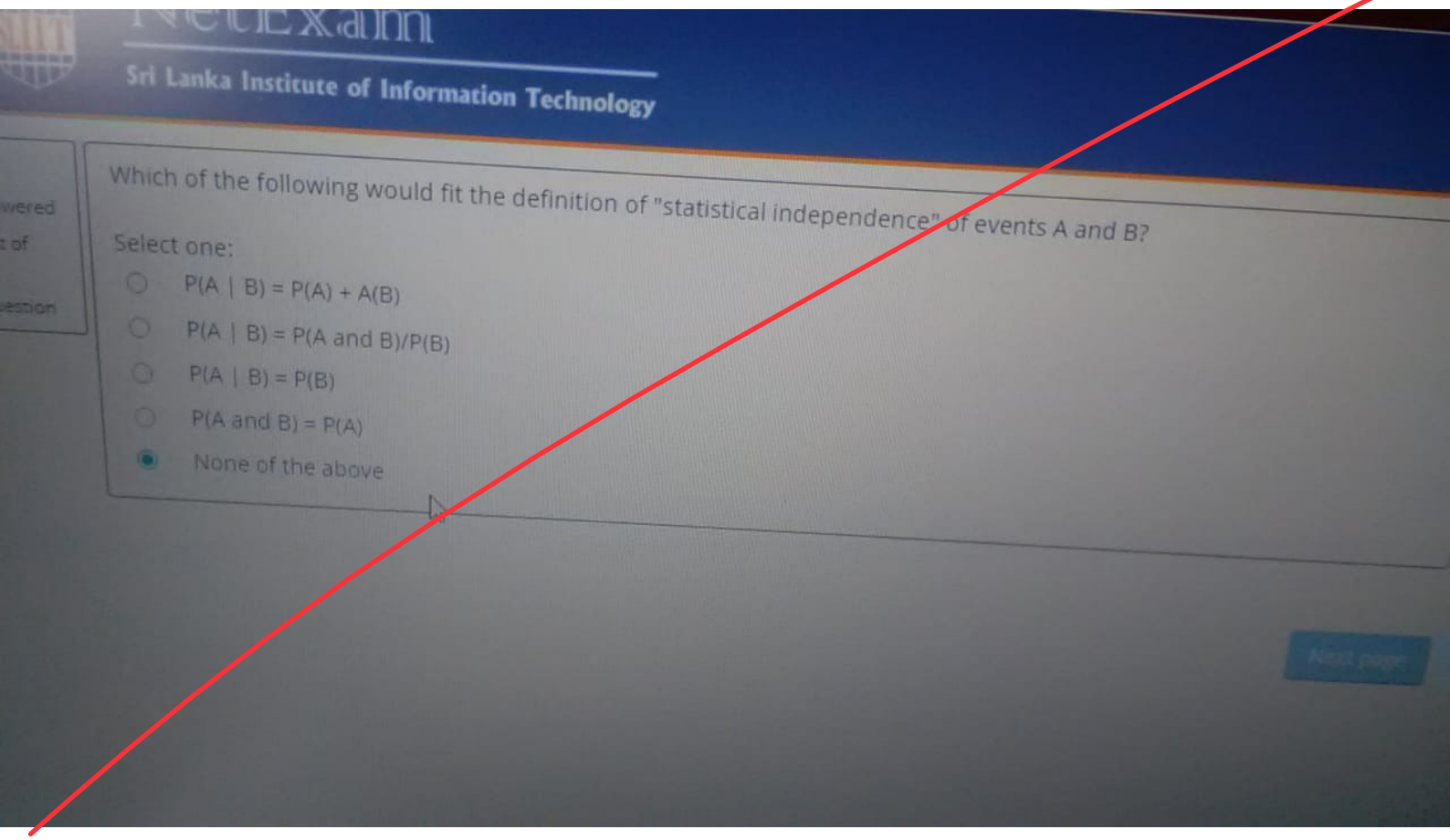
A machine produces parts that are either good (90%), slightly defective (2%), or obviously defective (8%). Produced parts get passed through an automatic inspection machine, which is able to detect any part that is obviously defective and discard it. The machine passes the part if it is not obviously defective. If it makes through the inspection machine and get shipped, what is the probability that it is a good quality part?

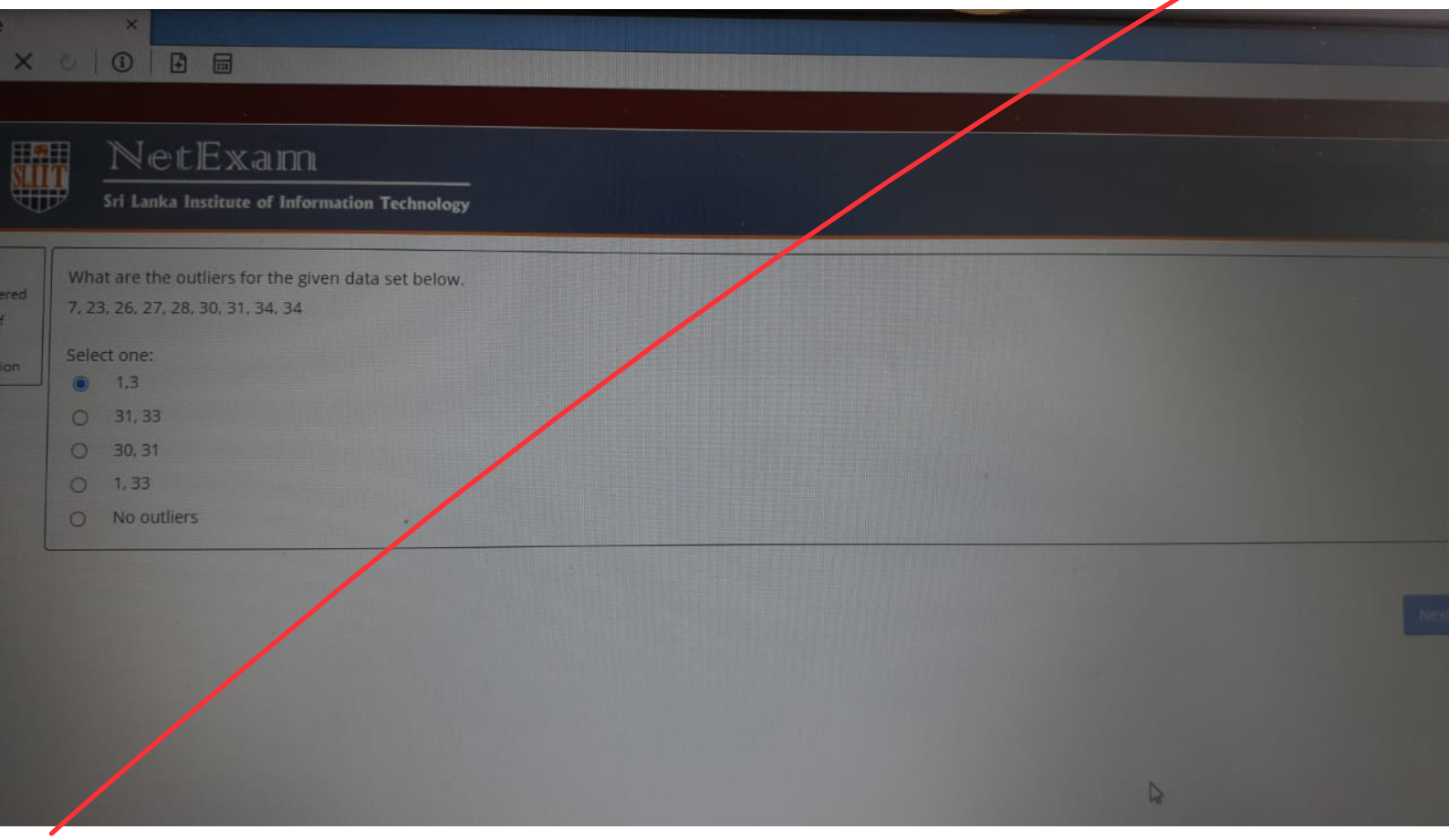
Select one:

- 0.9
- 0.978
- 0.742
- 0.685
- None of the above

Next page


NetExam
Sri Lanka Institute of Information Technology
Let E be an event and E' is its complement. If P(E) = 1/3, what is P(E')?
Select one:
P(E') = P(E) = 1/3
P(E') = P(E) - 1 = -2/3
P(E') =2* P(E) = 2/3
P(E') = 1- P(E) = 2/3
None of the above


Sri Lanka Institute of Information Technology
Which of the following would fit the definition of "statistical independence" of events A and B?
Select one:
P(A | B) = P(A) + A(B)
P(A | B) = P(A and B)/P(B)
P(A | B) = P(B)
P(A and B) = P(A)
None of the above


NetExam
Sri Lanka Institute of Information Technology
What are the outliers for the given data set below.
7, 23, 26, 27, 28, 30, 31, 34, 34
Select one:
1,3
31, 33
30, 31
1, 33
No outliers

NetExam

Sri Lanka Institute of Information Technology

Find E(X) for the random variable X with table:

values of X:	1	3	5
P(X=x):	1/6	1/6	2/3

Select one:

- 4
- 1
- 10
- 8
- None of the above

NetExam

Sri Lanka Institute of Information Technology

The probability of a machine producing a defective part is 0.04.

What is the probability of having exactly 5 defective parts, if it is selected from a sample of 100? (Round your answer upto 5 decimal places)

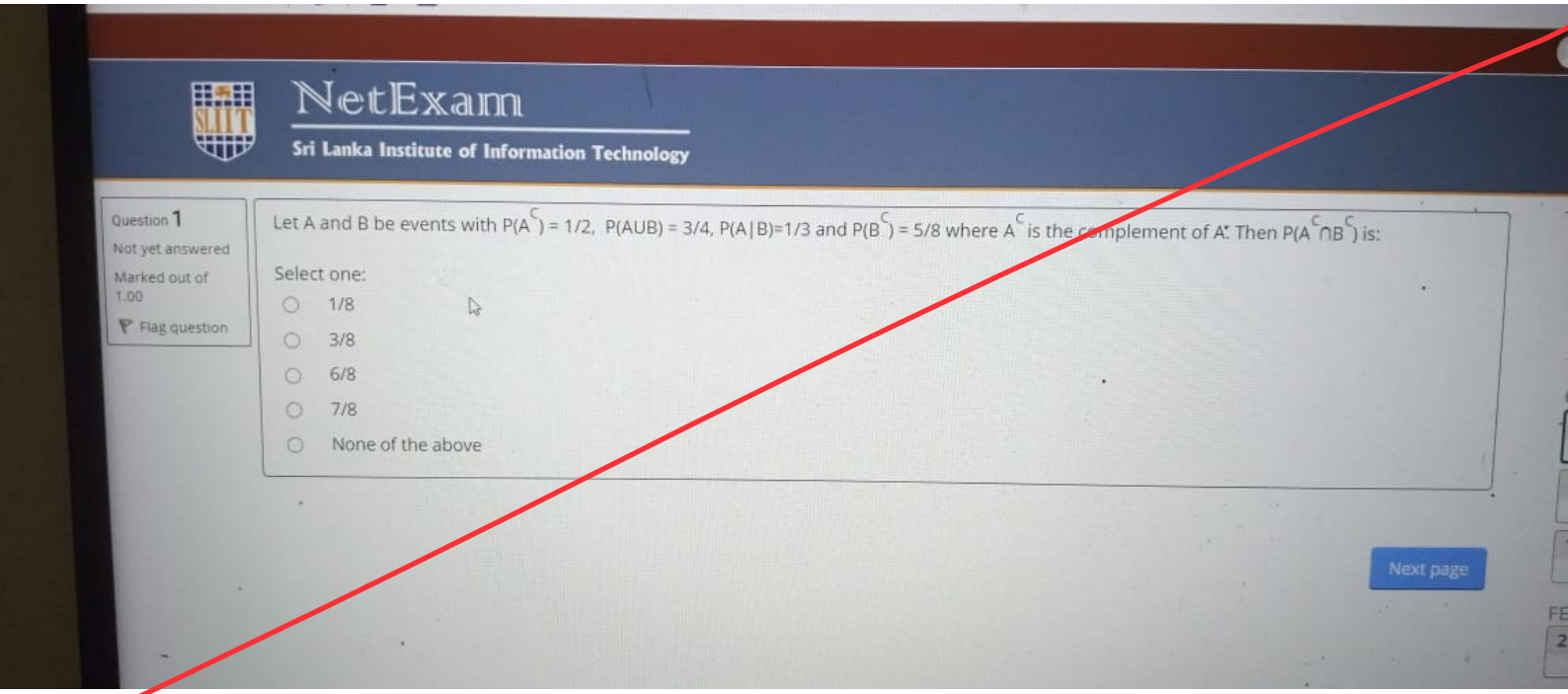
Choose...

Choose...
0.36926
0.15629
0.47116
3.84
4

What is the rate of the occurrence in this distribution?

What is the variance of this distribution?

Next page


NetExam
Sri Lanka Institute of Information Technology
Question 1
Not yet answered
Marked out of 1.00
Flag question
Let A and B be events with P(A^C) = 1/2, P(AUB) = 3/4, P(A|B)=1/3 and P(B^C) = 5/8 where A^C is the complement of A. Then P(A^C ∩ B^C) is:
Select one:
1/8
3/8
6/8
7/8
None of the above
Next page

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To be an outlier for the following data set, data points should lie between

18 44 47 55 61 62 78 79 83 145

Hint: Outlier is defined as a value x, holding the following conditions.

x > Q3 + 1.5*(Inter Quartile Range)

x < Q1 - 1.5*(Inter Quartile Range)

Lower bound -4.375

Upper bound 130.625

Next page

NetExam

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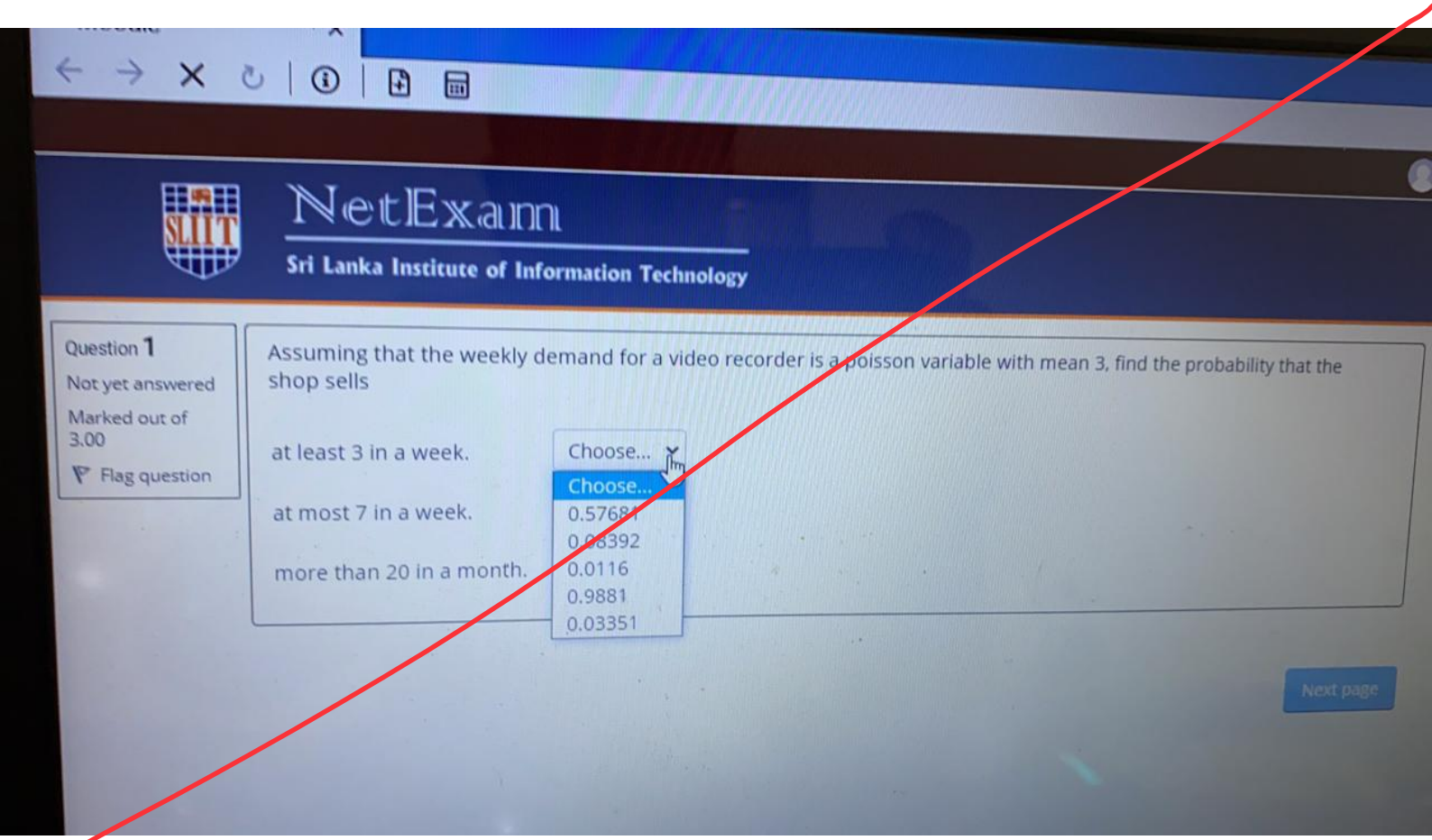
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A coin is tossed and a single 6-sided die is rolled. What is the probability of landing on the head side of the coin and rolling a 3 on the die?

Select one:

- 1/3
- 1/6
- 1/2
- 1/12
- None of the above

Next pa


NetExam
Sri Lanka Institute of Information Technology
Question 1
Not yet answered
Marked out of 3.00
Flag question
Assuming that the weekly demand for a video recorder is a poisson variable with mean 3, find the probability that the shop sells
at least 3 in a week.
at most 7 in a week.
more than 20 in a month.
Choose...
Choose...
0.57681
0.08392
0.0116
0.9881
0.03351
Next page

NetExam

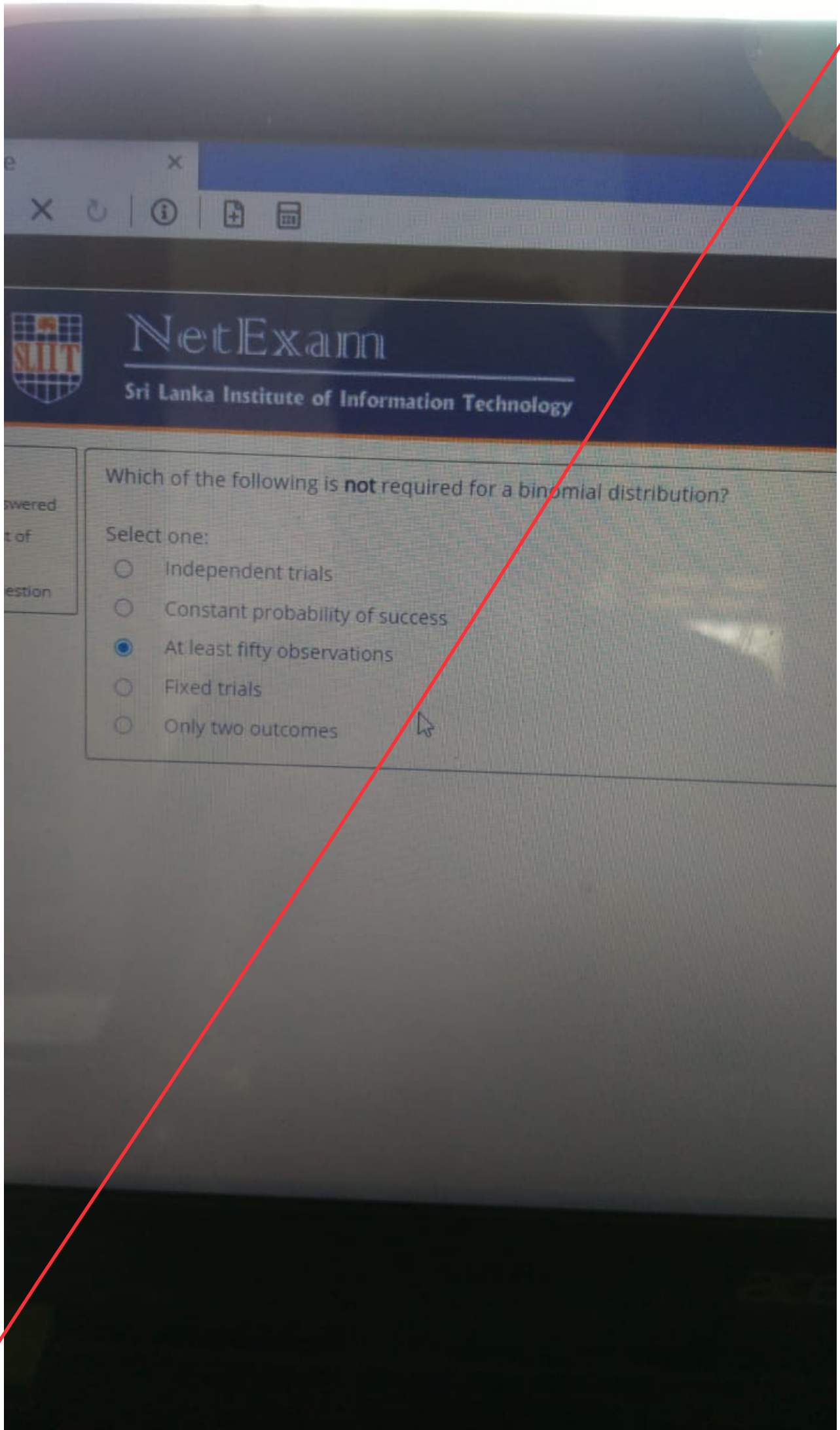
Sri Lanka Institute of Information Technology

The probability that a patient recovers from a heart operation is 0.9 What is the probability that at least 2 of the next three patients who have this operation recover? (Don't use any approximation)

Select one:

- 0.8960
- 0.9720
- 0.7890
- 0.5960
- None of the above

Next page


NetExamn
Sri Lanka Institute of Information Technology
Which of the following is not required for a binomial distribution?
Select one:
Independent trials
Constant probability of success
At least fifty observations
Fixed trials
Only two outcomes

NetExamn

Sri Lanka Institute of Information Technology

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estion

Which of the following can be a probability function?

Select one:

- P(X)=1/2 for x=1,2,3
- P(x)=x/5 for x=1,2,3,4
- P(X)=x/2 for x=0,1,2
- P(x)=1/5 for x=0,2,3
- None of the above

NetExam

Sri Lanka Institute of Information Technology

Determine the value of k so that the function $P(X=x)= kx$ for $x=1,2,3,4,5,6$ can serve discrete random variable X.

Select one:

- 3/21
- 5/21
- 1/21
- 2/21
- None of the above

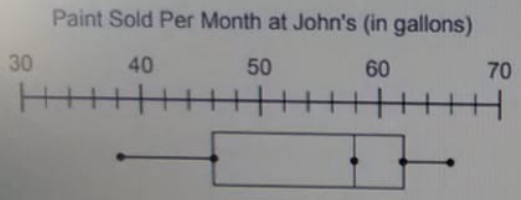
Question 4

Not yet answered

Marked out of 1.00

Flag question

According to the box-and-whisker plot, what is the third quartile of gallons of paint sold at John's Hardware Store in a month?



Select one:

- 66
- 61
- 60
- 62
- 58

Moodle

NetExam

Sri Lanka Institute of Information Technology

Question 3

Not yet answered

Marked out of 1.00

Flag question

A medical treatment has a success rate of 0.8. Two patients will be treated with this treatment. Assuming the results are independent for the two patients, what is the probability that neither one of them will be successfully cured?

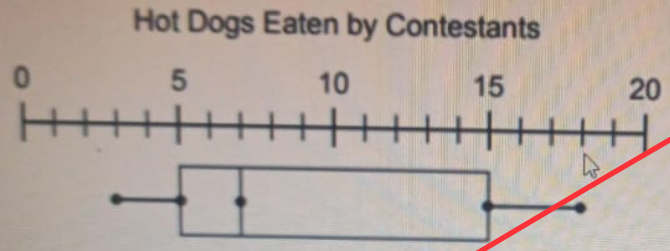
Select one:

- 0.5
- 0.36
- 0.2
- 0.04
- None of the above

Next page

n 6
t answered
d out of
g question

According to the box-and-whisker plot, what is the maximum number of hot dogs eaten in the hot dog eating contest?



Select one:

- 20
- 18
- 7
- [illegible]
- 15

Sri Lanka Institute of Information Technology

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estion

A and B are two events. P(A and B)' is equal to;

Select one:

- P(A' or B)
- P(A and B')
- P(A' and B')
- P(A' or B')
- None of the above.

Sri Lanka Institute of Information Technology

Question 4

Not yet answered

Marked out of 1.00

Flag question

The best sample is one that is,

Select one:

- A systematic sample
- Representative of the population
- Convenient
- judgmentally selected
- Non-random sample

NetExam

Sri Lanka Institute of Information Technology

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Flag question

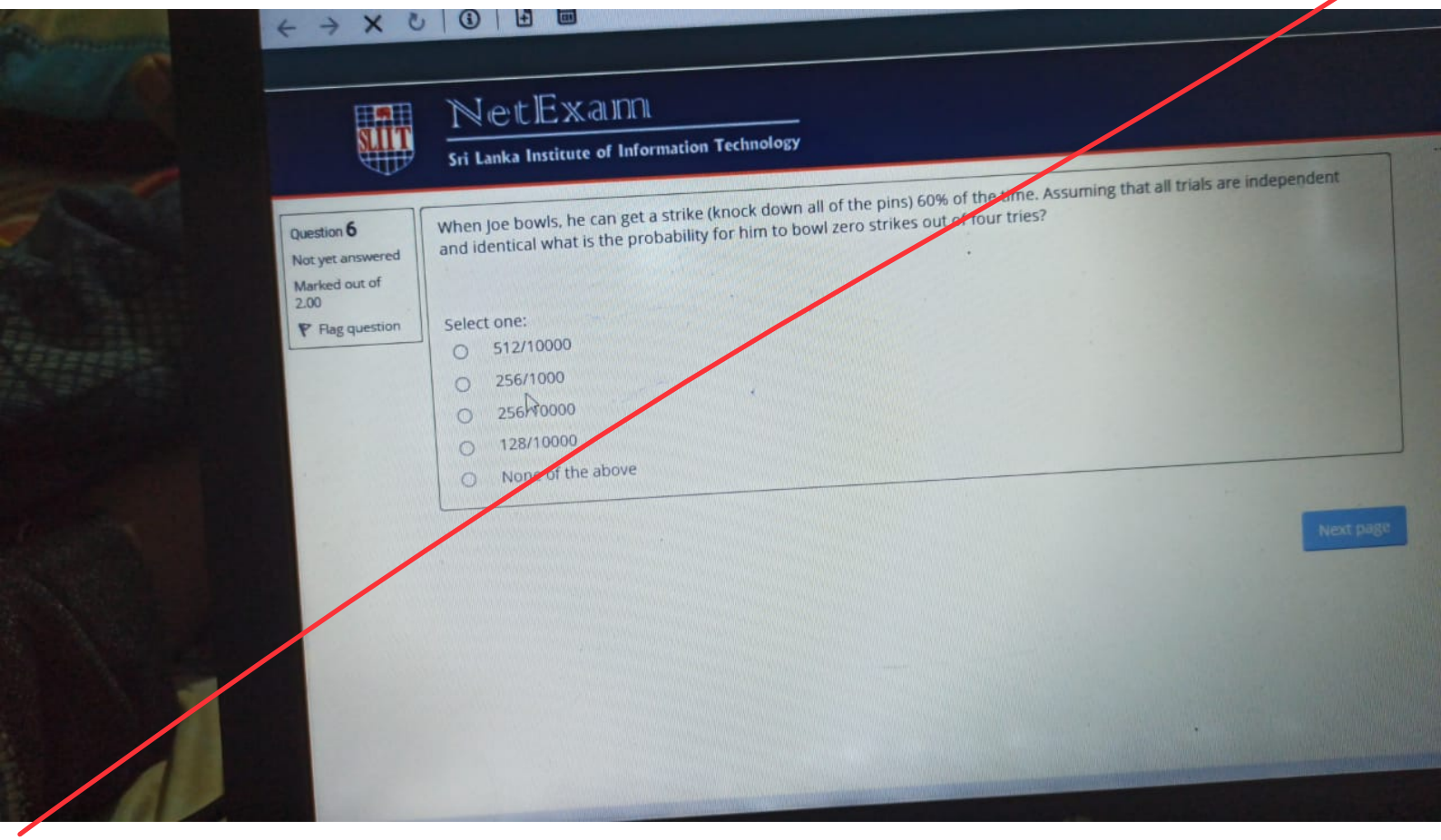
The below transformation is applied to a data set

$Y = 3X + 4$

Where X is old data and Y is New data. If the variance of the old

Select one:

- 49
- 5
- 9
- 45
- None of the above


NetExam
Sri Lanka Institute of Information Technology
Question 6
Not yet answered
Marked out of 2.00
Flag question
When Joe bowls, he can get a strike (knock down all of the pins) 60% of the time. Assuming that all trials are independent and identical what is the probability for him to bowl zero strikes out of four tries?
Select one:
512/10000
256/1000
256/10000
128/10000
None of the above
Next page

Sri Lanka Institute of Information Technology

The class midpoint is

Select one:

- The center of the class
- The lower limit of the class
- The upper limit of the class
- The width of the class
- The number of observations in a class

NetExam

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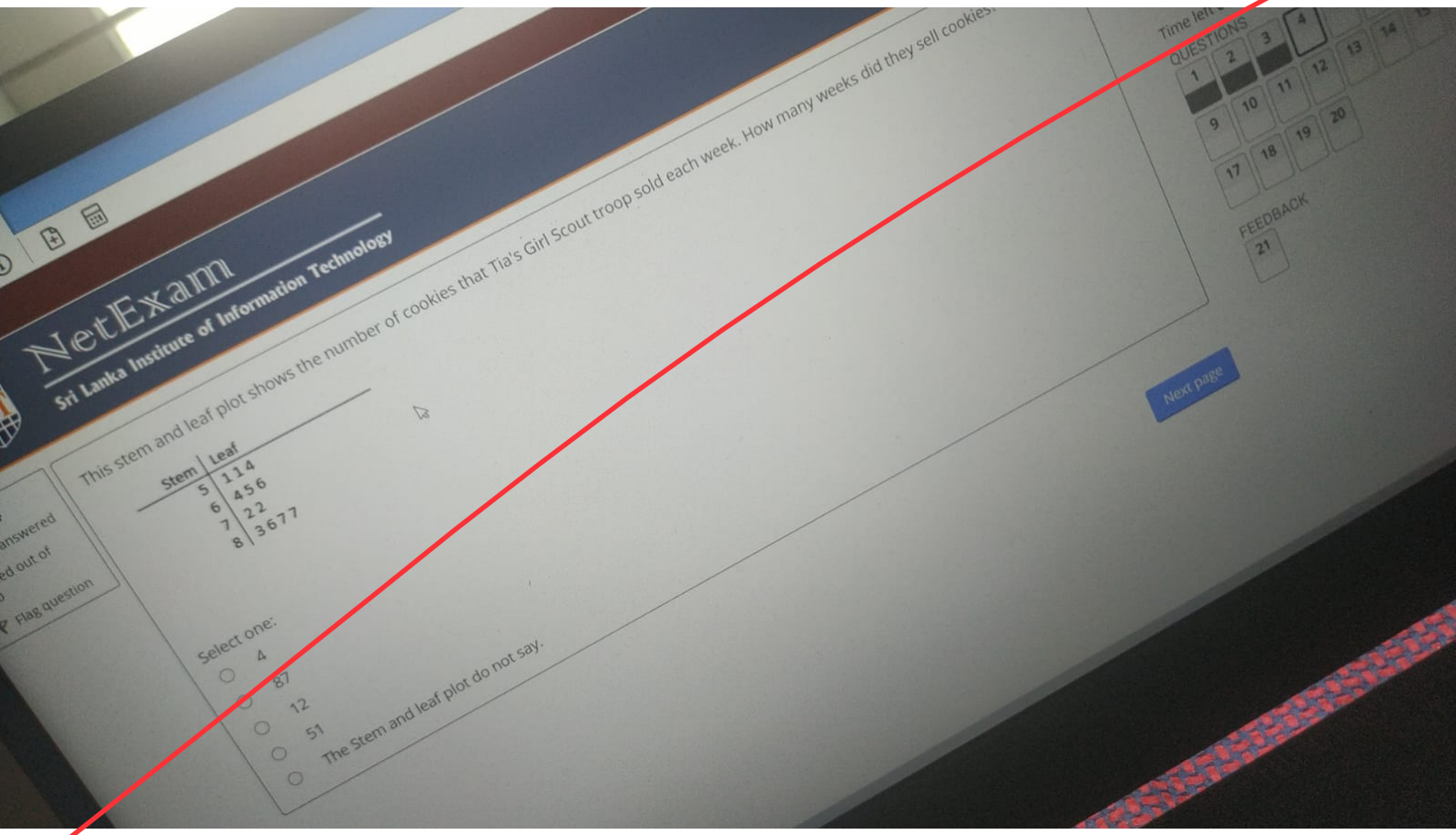
The below transformation is applied to a data set.

Y = 2X + 3, where X is the old data and Y is the new data.

If the variance of the first data set is 7 then what is the variance of the new data set.

Select one:

- 24
- 28
- 14
- 7
- None of the above


NetExam
Sri Lanka Institute of Information Technology
This stem and leaf plot shows the number of cookies that Tia's Girl Scout troop sold each week. How many weeks did they sell cookies
Stem | Leaf
5 | 1 1 4
6 | 4 5 6
7 | 2 2
8 | 3 6 7 7
Select one:
4
87
12
51
The Stem and leaf plot do not say.
Flag question
QUESTIONS
FEEDBACK
Next page

NetExam

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What are the outliers for the given data set below. It found that Q1 = 24.5, Q2 = 28 and Q3 = 32.5 for the data set.

7, 23, 26, 27, 28, 30, 31, 34, 34

Select one:

- 23
- 7
- 34
- 26
- No outliers

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NetExam

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A study is under way in to determine the adult height of American pine trees. Specifically, the study is attempting to determine what factors aid a tree in reaching heights greater than 60 feet tall. It is estimated that the forest contains 25,000 adult American pines. The study involves collecting heights from 250 randomly selected adult American pine trees and analyzing the results. Identify the sample in the study.

Select one:

- The 250 randomly selected adult American pine trees.
- The 25,000 adult American pine trees in the forest.
- All the adult American pine trees taller than 60 feet.
- All American pine trees, of any age, in the forest.
- None of the above.

Next pa

NetExam

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it20236182 Bandara P...

The probability that a patient recovers from a heart operation is 0.9 What is the probability that at least 2 of the next three patients who have this operation recover? (Don't use any approximation)

Select one:

- 0.8960
- 0.9720
- 0.7890
- 0.5960
- None of the above

Next page

Quiz navigation

Finish attempt ...

Time left **0:49:01**

QUESTIONS

1	2	3	4	5
9	10	11	12	13
17	18	19	20	

FEEDBACK

21

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The type of sampling in which each member of the population selected for the sample is returned to the population before the next member is selected is called,

Select one:

- Sampling without replacement
- Sampling with replacement
- Simple random sampling
- Systematic sampling
- None of the above

Next page

Moodle

NetExam

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Question 2

Not yet answered

Marked out of 2.00

Flag question

Which of the following is **not** required for a binomial distribution?

Select one:

- Independent trials
- At least fifty observations
- Fixed trials
- Only two outcomes
- Constant probability of success

on 7
et answered
ed out of
g question

This stem and leaf plot shows the scores of students for a test given out of 100. How many students scored more than 75 marks?

Stem	Leaf
5	2, 6, 6, 8
6	0, 1, 4, 9
7	1, 1, 2, 4, 6, 6, 7
8	0, 0, 3, 4, 5, 5, 7, 8, 9
9	0, 1, 1, 2, 6, 9

Select one:

- 12
- 2
- 6
- 18
- 8

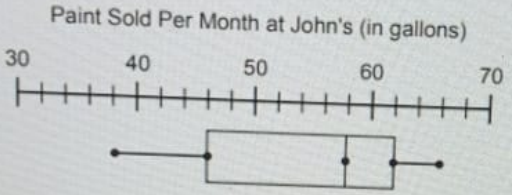
Question 4

Not yet answered

Marked out of 1.00

Flag question

According to the box-and-whisker plot, what is the third quartile of gallons of paint sold at John's Hardware Store i



Select one:

- 60
- 61
- 66
- 62
- 58

What are the outliers for the given data set below.

7, 23, 26, 27, 28, 30, 31, 34, 34

Select one:

- 1,3
- 31, 33
- 30, 31
- 1, 33
- No outliers

NetExam

Sri Lanka Institute of Information Technology

A numerical value used as a summary measure for a sample, such as sample mean, is known as a,

Select one:

- Population parameter
- Sample parameter
- Sample statistic
- Population mean
- None of the above

NetExam

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Question 10

Not yet answered

Marked out of 1.00

Flag question

Suppose in a town three candidates are running for the mayor's seat. To predict the winning candidate, a newspaper selects 25 registered voters randomly and the following info about each respondents' choice.

C C A B C C B C B B A B C A B A A C C A B A B C C

Choose the correct statement.

Select one:

- Population size is 25
- Sample size is not given
- Population is all registered voters in the town at the time of the study.
- Cannot identify a variable from the description
- None of the above

Next page

Quiz navigation

Finish attempt ...

Time left 0:34:39

QUESTIONS

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	

FEEDBACK

21

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Sri Lanka Institute of Information Technology

Question 9

Not yet answered

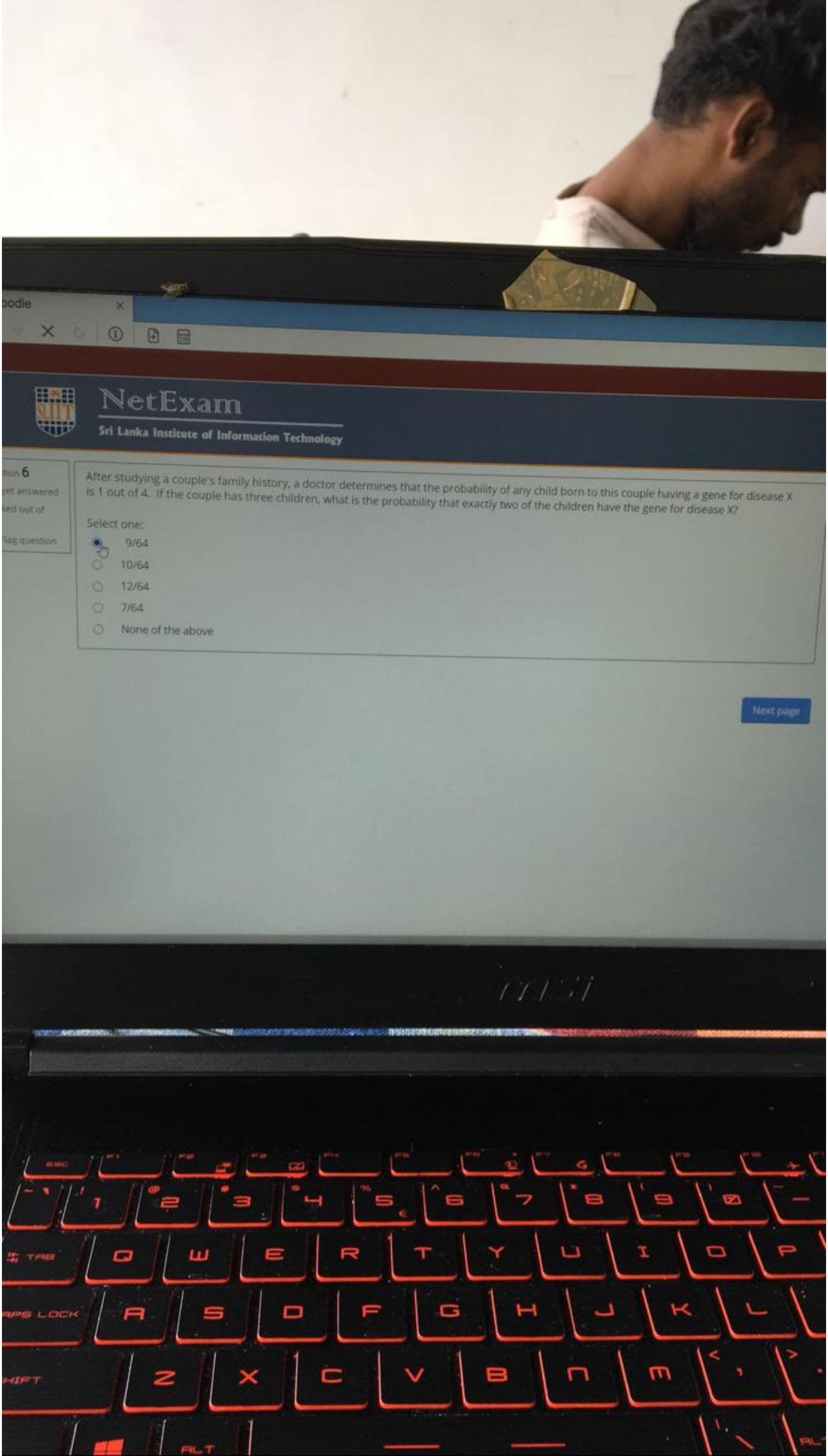
Marked out of 1.00

Flag question

The tallest bar in a histogram represents?

Select one:

- The class with the highest cumulative frequency
- The class with the lowest relative frequency
- The class with the highest frequency
- The class with the lowest frequency
- None of the above


NetExam
Sri Lanka Institute of Information Technology
After studying a couple's family history, a doctor determines that the probability of any child born to this couple having a gene for disease X is 1 out of 4. If the couple has three children, what is the probability that exactly two of the children have the gene for disease X?
Select one:
9/64
10/64
12/64
7/64
None of the above
Next page

NetExam

Sri Lanka Institute of Information Technology

A garment factory produces large lots of a certain type of garments. What is the mean of the number of defective units found in a sample of 10 units if the lot is 2% defective?

Select one:

- 5
- 2
- 0.2
- 0.98
- None of the above

NetExamn

Sri Lanka Institute of Information Technology

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When each member of a population has an equally likely chance of being selected, this is called:

Select one:

- A nonrandom sampling method.
- A quota sample.
- A judgement sampling
- A simple random sample.
- None of the above.

NetExam

Sri Lanka Institute of Information Technology

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Let E be an event and E' is its complement. If $P(E) = 1/3$, what is $P(E')$?

Select one:

- $P(E') = P(E) = 1/3$
- $P(E') = P(E) - 1 = -2/3$
- $P(E') = 2^* P(E) = 2/3$
- $P(E') = 1 - P(E) = 2/3$
- None of the above

Roll a fair die twice. Let X be the random variable that gives the absolute value of the differences between the two numbers.

X = | value of roll 1 – value of roll 2 |

Then what is the P(X=1)?

Select one:

- 1/6
- 10/36
- 8/36
- 9/36
- 2/36

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NetExam

Sri Lanka Institute of Information Technology

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question

In an experiment of rolling two dice, the first die shows a ONE and the other die rolls under the table and you cannot see it. Now, what is the probability that both die show ONE?

Select one:

- 1/3
- 1/6
- 1/36
- 9/36
- None of the above

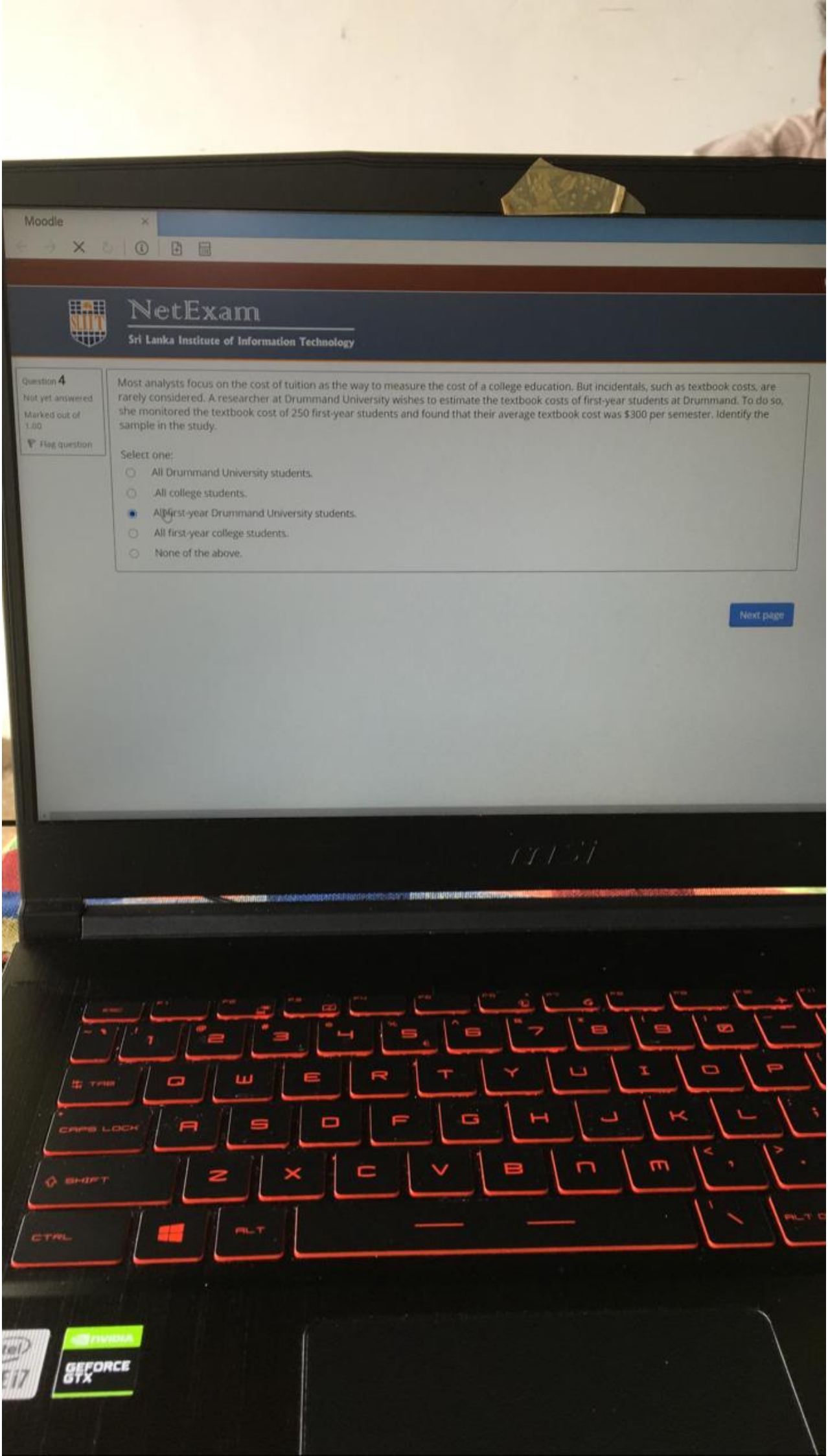
NetExam

Sri Lanka Institute of Information Technology

The class midpoint is

Select one:

- The center of the class
- The width of the class
- The upper limit of the class
- The number of observations in a class
- The lower limit of the class


Moodle
NetExam
Sri Lanka Institute of Information Technology
Question 4
Not yet answered
Marked out of 1.00
Flag question
Most analysts focus on the cost of tuition as the way to measure the cost of a college education. But incidentals, such as textbook costs, are rarely considered. A researcher at Drummand University wishes to estimate the textbook costs of first-year students at Drummand. To do so, she monitored the textbook cost of 250 first-year students and found that their average textbook cost was $300 per semester. Identify the sample in the study.
Select one:
All Drummand University students.
All college students.
All first-year Drummand University students.
All first-year college students.
None of the above.
Next page
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GEFORCE GTX

A and B are two events. P(A and B)' is equal to;

Select one:

- P(A' or B)
- P(A and B')
- P(A' and B')
- P(A' or B')
- None of the above.

Next page

NetExam

Sri Lanka Institute of Information Technology

Question 2

Not yet answered

Marked out of 1.00

Flag question

Which of the following is not a form of non-probability sampling?

Select one:

- Quota sampling.
- Convenience sampling.
- Cluster sampling.
- Purposive/Judgement sampling.
- They are all forms of non-probability sampling.

Question 1

Not yet answered

Marked out of 1.00

Flag question

The conditional probability of x given y is:

Select one:

- the probability that x and y occur jointly
- the probability that y occurs if x has already occurred
- the probability that x occurs if y has already occurred
- the marginal probability of x minus the marginal probability of y
- None of the above

Next page

NetExam

Sri Lanka Institute of Information Technology

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Sampling in which a sampling unit can be repeated more than once is called,

Select one:

- Sampling without replacement
- Simple sampling
- Sampling with replacement
- Repeated sampling
- None of the above

Consider the following probability function

$P(X=x)=cx^2$; x=3,4,5, where c is positive constant. Find c.

Select one:

- 0.2
- 0.2
- 0.02
- 2
- None of the above

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Your neighbor has 2 children. You learn that he has a son, Joe. What is the probability that Joe's sibling is a brother? (Assume that boys and girls are equally likely).

Select one:

- 1/4
- 1/5
- 1/3
- 1/2
- None of the above

X	-10	-20	30
P(X)	1/5	3/10	1/2

Let X be a random variable with the probability distribution given above. The mean of $g(X) = 2X$ is:

Select one:

- 1
- 7
- 14
- -1
- None of the above

Consider the following discrete probability distribution for the random variable X.

X	0	1	2
$P(X=x)$	a	b	0.4

If the mean of X is 1 then,

Select one:

- $a=0.3$ and $b=0.1$
- $a=0.2$ and $b=0.4$
- $a=0.4$ and $b=0.2$
- $a=0.2$ and $b=0.2$
- $a=0.1$ and $b=0.5$

Next page

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Question 3

Not yet answered

Marked out of 1.00

Flag question

A medical treatment has a success rate of 0.8. Two patients will be treated with th
two patients, what is the probability that neither one of them will be successfully

Select one:

- 0.5
- 0.36
- 0.2
- 0.04
- None of the above

NetExam

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If the number of arrivals in a queue is 10 per hour on average, determine the probability that, in any hour there will be

0 arrivals Choose...

6 arrivals Choose...

more than 6 arrivals Choose...

Next page

Quiz n

Finish attempt

Time left 0:27:5

QUESTIONS

1 2 3

9 10 11

17 18 19

FEEDBACK

21

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μ is an example of a

Select one:

- population parameter
- sample statistic
- population variance
- sample variance
- None of the above

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Question 5

Not yet answered

Marked out of 3.00

Flag question

In a district, the probability of having a power cut in a house will be 0.003.

What is the probability that 10 houses having the power cut, out of 1000 houses in this district? (Give your answer up to 5 decimal places)

What is the variance of this distribution?

What is the rate of the occurrence in this distribution?

Choose...

Choose...
0.20018
0.00081
4.5
3
0.61110

Next page

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Question 12

Not yet answered

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Flag question

Consider the following discrete probability distribution for the random variable X.

X	1	2	3	4	5
$P(X=x)$	p	$2p$	$3p$	$4p$	$5p$

The mean of X is,

Select one:

- 2
- 3
- 3.5
- 3.67
- 5.21

Next page

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Which of the following is **not** required for a binomial distribution?

Select one:

- Fixed trials
- Constant probability of success
- Independent trials
- Only two outcomes
- At least fifty observations

NetExam

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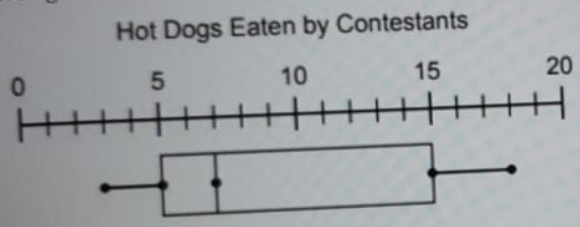
Question 4

Not yet answered

Marked out of 1.00

Flag question

According to the box-and-whisker plot, what is the maximum number of hot dogs eaten in the hot dog eating contest?



Select one:

- 18
- 7
- 3
- 20
- 15

Next page

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Question 1

Not yet answered

Marked out of 2.00

Flag question

Consider the following probability

$P(X=x)=(1/12)* x$, for $x=3,4,5$

Find the expected value

Select one:

- 3
- 0.3
- 4.28
- 2.5
- None of the above

Next page

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a district, the probability of having a power cut in a house will be 0.0015.

hat is the probability that at most 12 houses having the power cut, out of 3000 houses in the district? (Give your answer to 5 decimal places) 0.99919

hat is the variance of this distribution? 4.5

hat is the rate of the occurrence in this distribution? 0.00240

Next page

NetExam

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g question

P(A)=3/8,P(B)=5/8,and P(A∪B)=3/4, Find P(A∩B)

Select one:

- 1/5
- 1/4
- 1/8
- 3/8
- None of the above

NetExam

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A numerical value used as a summary measure for a sample, such as sample mean, is known as a,

Select one:

- Population parameter
- Sample parameter
- Sample statistic
- Population mean
- None of the above

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NetExam

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Flag question

This stem and leaf plot shows the number of cookies that Tia's Girl Scout troop sold each week. How many weeks did they

Stem	Leaf
5	1 1 4
6	4 5 6
7	2 2
8	3 6 7 7

Select one:

- 4
- 87
- 12
- 51
- The Stem and leaf plot do not say.

This stem and leaf plot shows the scores of students for a test given out of 100. How many students scored more than 75 marks?

Stem	Leaf
5	2, 6, 6, 8
6	0, 1, 4, 9
7	1, 1, 2, 4, 6, 6, 7
8	0, 0, 3, 4, 5, 5, 7, 8, 9
9	0, 1, 1, 2, 6, 9

Select one:

- 6
- 18
- 8
- 12
- 2

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question

Which of the following is a discrete quantitative variable?.

Select one:

- The Dow Jones Industrial (stock market) average .
- The volume of water released from a dam.
- The distance you drove yesterday.
- The number of employees of an insurance company
- None of the above.

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Which of the following is **not** required for a binomial distribution?

Select one:

- Constant probability of success
- At least fifty observations
- Fixed trials
- Only two outcomes
- Independent trials

NetExam

Sri Lanka Institute of Information Technology

Find E(X) for the random variable X with table:

values of X:	1	3	5
P(X=x):	1/6	1/6	2/3

Select one:

- (●) 4
- ( ) 1
- ( ) 10
- ( ) 8
- ( ) None of the above

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Consider the following probability

$P(X=x)=(1/12)^* x$, for $x=3,4,5$

Find the expected value

Select one:

- 3
- 0.3
- 4.28
- 2.5
- None of the above

NetExam

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Which of the following is a discrete quantitative variable?.

Select one:

- The Dow Jones Industrial (stock market) average .
- The volume of water released from a dam.
- The distance you drove yesterday.
- The number of employees of an insurance company
- None of the above.

NetExamn

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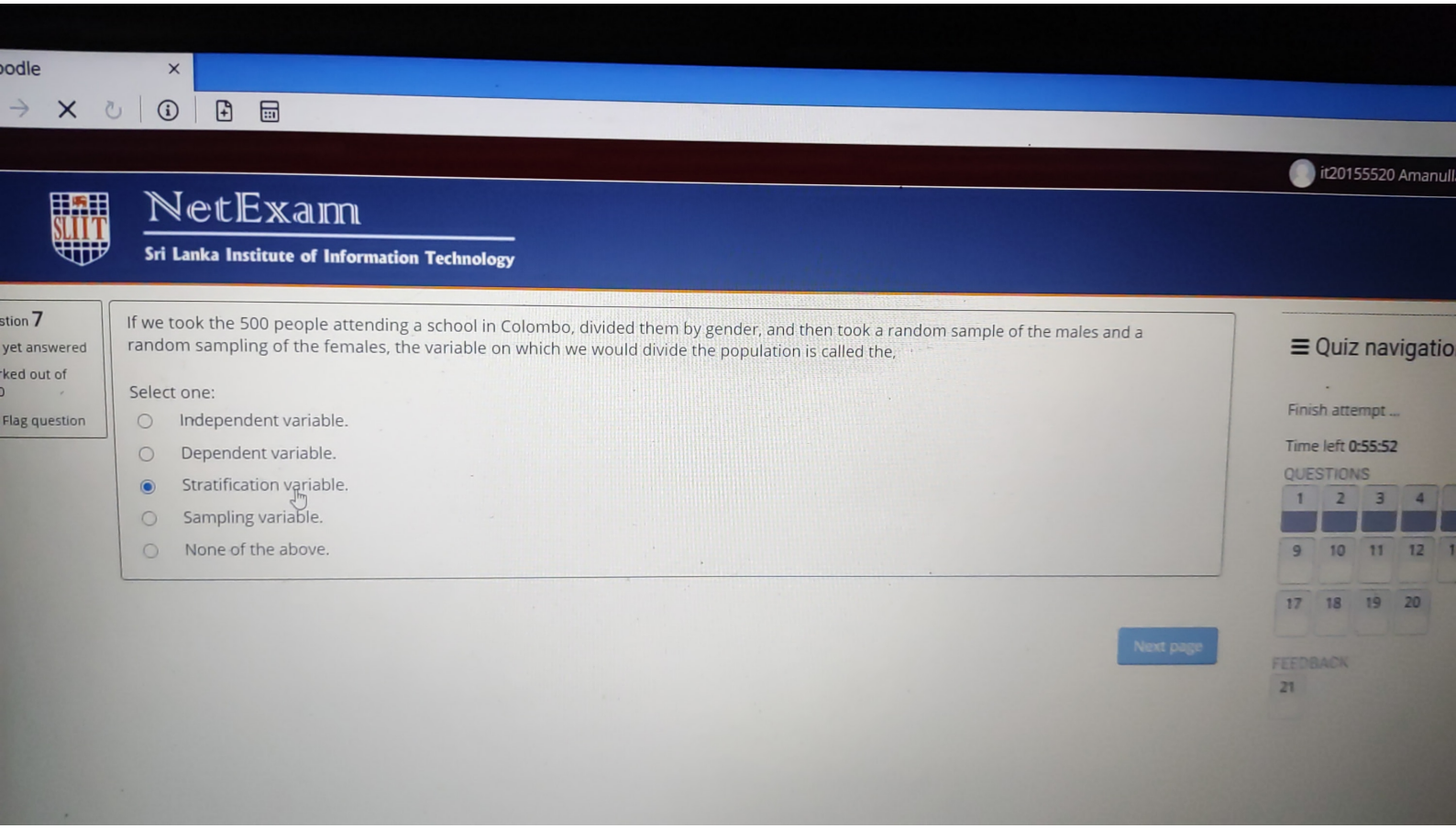
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The class midpoint is

Select one:

- The width of the class
- The upper limit of the class
- The number of observations in a class
- The center of the class
- The lower limit of the class

Next page


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NetExam
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Flag question
If we took the 500 people attending a school in Colombo, divided them by gender, and then took a random sample of the males and a random sampling of the females, the variable on which we would divide the population is called the,
Select one:
Independent variable.
Dependent variable.
Stratification variable.
Sampling variable.
None of the above.
Next page
Quiz navigatio
Finish attempt ...
Time left 0:55:52
QUESTIONS
1 2 3 4
9 10 11 12
17 18 19 20
FEEDBACK
21

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Flag question

X	-10	-20	30
P(X)	1/5	3/10	1/2

Let X be a random variable with the probability distribution given above. The mean of $g(X) = 2X$ is:

Select one:

- 1
- 7
- 14
- -1
- None of the above

NetExam

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Suppose 2% of the bolts produced by a factory are defective. In a shipment of 3600 bolts from the factory, find the mean number of defective bolts and the standard deviation without using any approximation.

Select one:

- Mean = 72, standard deviation = 70.56
- Mean = 72, standard deviation = 8.4
- Mean = 720, standard deviation = 24
- Mean = 720, standard deviation = 576
- None of the above

Next page

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Which of the following sampling techniques is an equal probability selection method (EPSEM) in which every individual in the population has an equal chance of being selected?

Select one:

- Simple random sampling.
- Systematic sampling.
- Proportional stratified sampling.
- Cluster sampling.
- All of the above are EPSEM.

Next page

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Suppose in a town three candidates are running for the mayor's seat. To predict the winning ca
voters randomly and the following info about each respondents' choice.

C C A B C C B C B B A B C A B A A C C A B A B C C

Choose the correct statement.

Select one:

- Population size is 25
- Sample size is not given
- Population is all registered voters in the town at the time of the study.
- Cannot identify a variable from the description
- None of the above

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Let A and B be events with $P(A^C) = 1/2$, $P(A \cup B) = 3/4$, $P(A|B) = 1/3$ and $P(B^C) = 5/8$ where A^C is the complement of A. Then $P(A^C \cap B^C)$ is:

Select one:

- 1/8
- 3/8
- 6/8
- 7/8
- None of the above

Next

NetExam

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g question

When arranging data into classes it is suggested that you have

Select one:

- Between 5 and 10 classes
- Always only 5 classes
- More than 20 classes
- Less than 5 classes
- Between 20 and 40 classes

The class midpoint is

Select one:

- The center of the class
- The width of the class
- The upper limit of the class
- The number of observations in a class
- The lower limit of the class

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A number calculated with complete population data and quantifies a characteristic of the population is called which of the following?

Select one:

- A datum
- A parameter
- A statistic
- A population
- None of the above

Sample mean and the median of the following data are respectively given by

-1, 2, 0, 3, 3, 4, 2

Select one:

- 2 and 1.875
- 1.8571 and 2
- 1.7209 and 1.9863
- 1.574 and 2
- None of the above

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What are the outliers for the given data set below.

7, 23, 26, 27, 28, 30, 31, 34, 34

Select one:

- 1,3
- 31, 33
- 30, 31
- 1, 33
- No outliers

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5

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question

In a district, the probability of having a power cut in a house will be 0.003.

What is the probability that 10 houses having the power cut, out of 1000 houses in this district? (Give your answer up to 5 decimal places)

What is the variance of this distribution?

What is the rate of the occurrence in this distribution?

Choose...

Choose...
0.20018
0.00081
4.5
3
0.61110

Next page

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Time left

QUESTION

1 2

9 10

17 18

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Consider the following probability function

$P(X=x)=cx^2$; x=3,4,5, where c is positive constant. Find c.

Select one:

- 0.2
- 0.2
- 0.02
- 2
- None of the above

NetExamn

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If the number of arrivals in a queue is 10 per hour on average, determine the probability that, in any hour there will be

0 arrivals Choose...

6 arrivals Choose...

more than 6 arrivals Choose...

Choose...
0.93291
0.06305
0.00005
0.86986

Question 8

Not yet answered

Marked out of 1.00

Flag question

Since the population size is always larger than the sample size, the sample statistic

Select one:

- can never be larger than the population parameter
- can never be equal to the population parameter
- some cases it can be equal to the population parameter
- can never be smaller than the population parameter
- None of the above

Next page

Most analysts focus on the cost of tuition as the way to measure the cost of a college education. But incidentals, such as textbook costs, are rarely considered. A researcher at Drummand University wishes to estimate the textbook costs of first-year students at Drummand. To do so, she monitored the textbook cost of 250 first-year students and found that their average textbook cost was $300 per semester. Identify the variable of interest to the researcher.

Select one:

- The textbook cost of first-year Drummand University students.
- The year in school of Drummand University students.
- The age of Drummand University students.
- The cost of incidental expenses of Drummand University students.
- None of the above.

Next page

Question 11

Not yet answered

Marked out of 1.00

Flag question

What is the probability that the sum of two die will be greater than 8, given that the first die is 6?

Select one:

- 1/2
- 3/4
- 2/3
- 7/12
- None of the above

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.00

Flag question

Select the correct answer from the below description

A survey conducted by a statistician interviewed 200 young men who didn't go to university. Of those who took restau jobs, one in two reached a higher level job and one in four reached a managerial position.

Select one:

- 200 young men are the population of interest
- The variable of interest is a qualitative variable.
- 25% is the population proportion of those who reached the managerial position.
- The sample size is not given in the description
- None of the above

Next p

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A dresser drawer contains one pair of socks with each of the following colours: blue, brown, red, white and black. Each pair is folded together in a matching set. You reach into the sock drawer and choose a pair of socks without looking. You replace this pair and then choose another pair of socks. What is the probability that you will choose the red pair of socks both times?

Select one:

- 1/5
- 1/25
- 1/10
- 1/15
- None of the above

Next page

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A personal computer user survey was conducted. Time of personal computer use per week is an example of a

Select one:

- Discrete numerical variable
- Continuous numerical variable
- Nominal categorical variable
- Ordinal categorical variable
- None of the above.

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The statistics course consists of two (2) online quizzes. 22% of the class passed both tests and 41% of the class passed the first test. About what percent of those who passed the first test also passed the second test? (Round up the answer to the nearest integer)

Select one:

- 54%
- 20%
- 3%
- 24%
- None of the above

Next page

Quiz na

Finish attempt

Time left 0:28:0

QUESTIONS

1 2 3

9 10 11

17 18 19

FEEDBACK

21

NetExam

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Question 15

Not yet answered

Marked out of 2.00

Flag question

If A is the event, "The team wins at least 5 foot ball games", then A' is;

Select one:

- The team wins more than 5 foot ball games
- The team wins less than 5 foot ball games
- The team does not win any foot ball game
- There is no sufficient information to answer the question
- None of the above

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Question 4

Not yet answered

Marked out of 3.00

Flag question

Assuming that the weekly demand for a video recorder is a poisson variable with mean 3, find the probability that the shop sells

at least 3 in a week. Choose...

at most 7 in a week.

more than 20 in a month.

Choose...
0.57681
0.9881
0.08392
0.03351
0.0116

Next page

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Let A and B be events with $P(A^C) = 1/2$, $P(A \cup B) = 3/4$, $P(A|B) = 1/3$ and $P(B^C) = 5/8$ where A^C is the complement of A. Then $P(A^C \cap B^C)$ is:

Select one:

- 1/8
- 3/8
- 6/8
- 7/8
- None of the above

Next page

NetExam

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What is the probability that the sum of two die will be greater than 8, given that the first die is 6?

Select one:

- 1/2
- 3/4
- 2/3
- 7/12
- None of the above

Let A and B be events with $P(A^C) = 1/2$, $P(A \cup B) = 3/4$, $P(A \mid B) = 1/3$ and $P(B^C) = 5/8$ where A^C is the complement of A. Then $P(A^C \cap B^C)$ is:

Select one:

- 1/8
- 3/8
- 6/8
- 7/8
- None of the above

Next page

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Question 13

Not yet answered

Marked out of 1.00

Flag question

At a certain school, 18% of all students play football and basketball and 32% of all students play football. What is the probability that a student plays basketball given that the student plays football? (Round up the answer to the nearest integer)

Select one:

- 56%
- 178%
- 50%
- 32%
- None of the above

Next page

Quiz navigation

Finish attempt ...

Time left 0:24:21

QUESTIONS

1	2	3	4	
8	9	10	11	
15	16	17	18	

FEEDBACK

21

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When Joe bowls, he can get a strike (knock down all of the pins) 60% of the time. Assuming that all trials are independent and identical what is the probability for him to bowl zero strikes out of four tries?

Select one:

- 512/10000
- 256/1000
- 256/10000
- 128/10000
- None of the above

NetExamn

Sri Lanka Institute of Information Technology

A garment factory produces large lots of a certain type of garments. What is the mean of the number of defective units found in a sample of 10 units if the lot is 2% defective?

Select one:

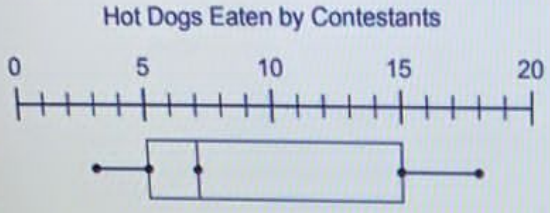
- 5
- 2
- 0.2
- 0.98
- None of the above

Next page

NetExam

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According to the box-and-whisker plot, what is the maximum number of hot dogs eaten in the hot dog eating contest?



Select one:

- 18
- 20
- 15
- 7
- 3

Next page

NetExam

Sri Lanka Institute of Information Technology

Question 15

Not yet answered

Marked out of 3.00

Flag question

If the number of arrivals in a queue is 10 per hour on average, determine the probability that, in any hour there will be

0 arrivals

6 arrivals

more than 6 arrivals

Choose...

Choose...
0.86986
0.00005
0.93291
0.06305

Next page

NetExam

Sri Lanka Institute of Information Technology

The probability of a machine producing a defective part is 0.02.

What is the probability of having more than 5 defective parts, out of a sample of 200? (Round your answer upto 5 decimal places)

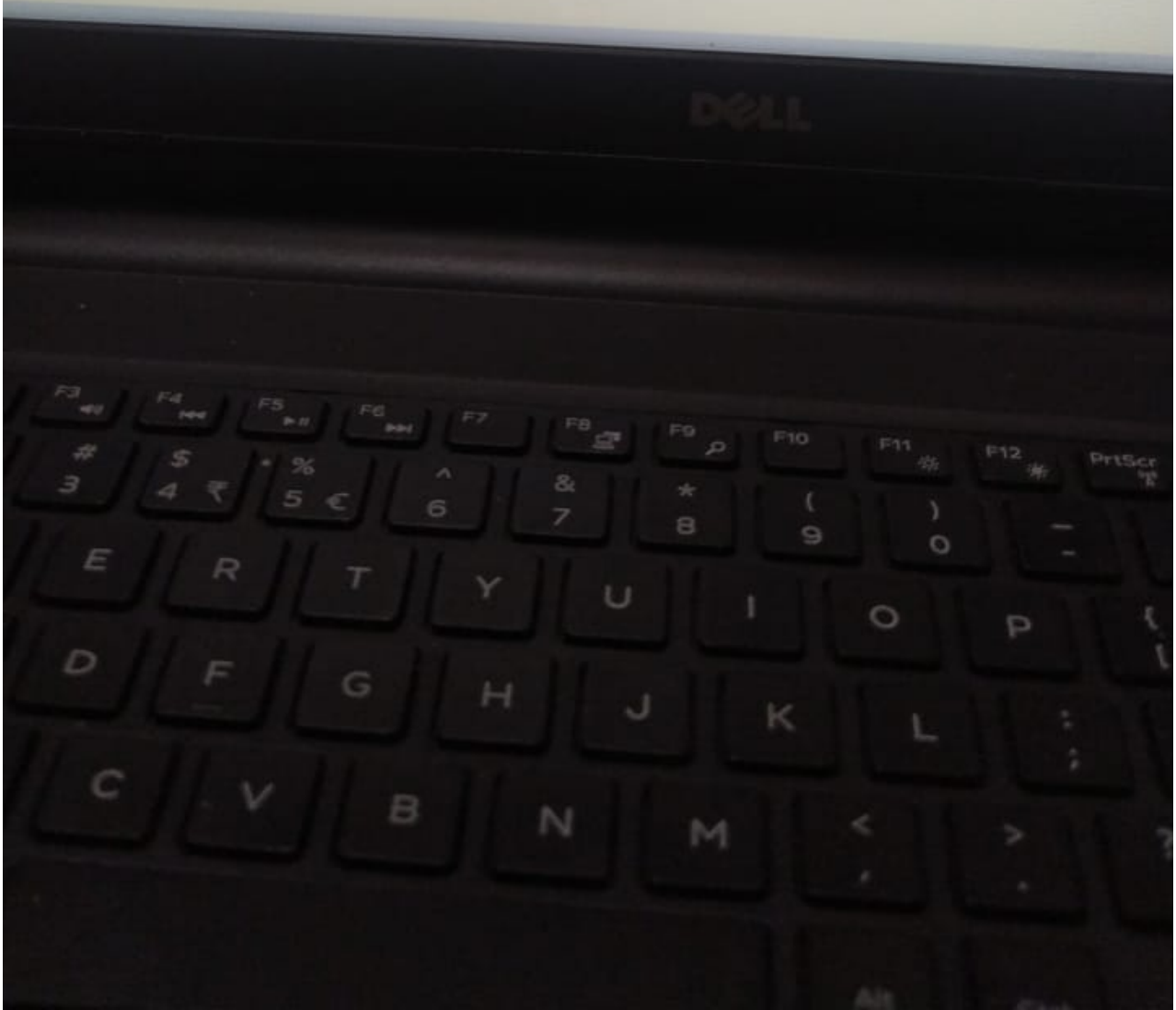
What is the rate of the occurrence in this distribution?

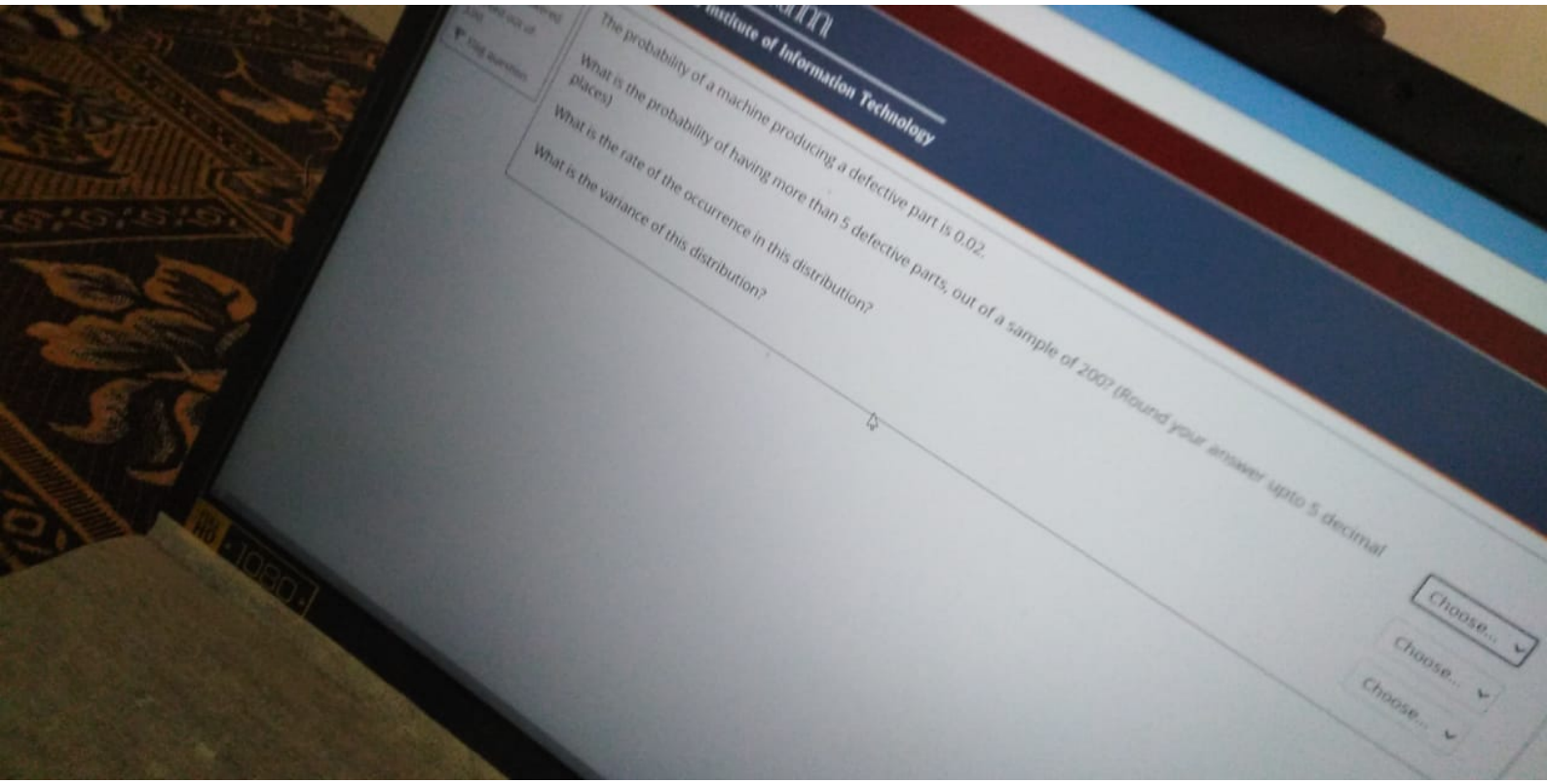
What is the variance of this distribution?

Choose...

Choose...
4
0.37116
2.92
0.41847
0.21487

Next p




Institute of Information Technology
The probability of a machine producing a defective part is 0.02.
What is the probability of having more than 5 defective parts, out of a sample of 200? (Round your answer upto 5 decimal places)
What is the rate of the occurrence in this distribution?
What is the variance of this distribution?
Choose...
Choose...
Choose...

NetExam

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Mode and the median of the following data set respectively

6, 5, 5, 4, 3, 5

Select one:

- 6 and 6
- 5 and 6
- 5 and 5.5
- 5 and 5
- None of the above

Question 6
Not yet answered
Marked out of 1.00
Flag question

The statistics course consists of two (2) online quizzes. 22% of the class passed both tests and 41% of the class passed the first test. About what percent of those who passed the first test also passed the second test? (Round up the answer to the nearest integer)

Select one:

- 54%
- 20%
- 3%
- 24%
- None of the above

Next page

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Let A and B be events with $P(A^C) = 1/2$, $P(A \cup B) = 3/4$, $P(A|B) = 1/3$ and $P(B^C) = 5/8$ where A^C is the complement of A. Then $P(A^C \cap B^C)$ is:

Select one:

- 1/8
- 3/8
- 6/8
- 7/8
- None of the above

Next page

NetExam

Sri Lanka Institute of Information Technology

A garment factory produces large lots of a certain type of garments. What is the mean of the number of defective units found in a sample of 10 units if the lot is 2% defective?

Select one:

- 5
- 2
- 0.2
- 0.98
- None of the above

Next page

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Question 18

Not yet answered

Marked out of 2.00

Flag question

Find the outliers, if any, for the following data set

18 44 47 55 61 62 78 79 83 145

Hint: Outlier is defined as a value x, holding the following conditions.

x > Q3 + 1.5*(Inter Quartile Range)

x < Q1 - 1.5*(Inter Quartile Range)

Is "18" an outlier: Yes

Is "145" an aoutlier?: No

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QUESTIO

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15 16

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NetExam

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The probability that a patient recovers from a heart operation is 0.9 What is the probability that at least 2 of the next three patients who this operation recover? (Don't use any approximation)

Select one:

- 0.8960
- 0.9720
- 0.7890
- 0.5960
- None of the above

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Determining the sample interval (represented by k), randomly selecting a number between 1 and k, and including each kth element in you sample are the steps for which form of sampling?

Select one:

- Simple Random Sampling
- Stratified Random Sampling
- Systematic Sampling
- Cluster sampling
- None of the above

Next

NetExam

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Assuming that the weekly demand for a video recorder is a poisson variable with mean 3, find the probability that the shop sells

at least 3 in a week. Choose...

at most 7 in a week.

more than 20 in a month.

Choose...
0.08392
0.9881
0.03351
0.57681
0.0116

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To be an outlier for the following data set, data points should lie between,

10.2, 14.1, 14.4, 14.4, 14.4, 14.5, 14.5, 14.6, 14.7, 14.7, 14.7, 14.9, 15.1, 15.9, 16.4

Hint: Outlier is defined as a value x, holding the following conditions.

x > Q3 + 1.5*(Inter Quartile Range)

x < Q1 - 1.5*(Inter Quartile Range)

Upper bound Choose...

Lower bound

- Choose...
- 15.25
- 13.15
- 15.35
- 16.15
- 13.65
- 13.45
- 15.45
- 12.45
- 15.65

Next page

NetExam

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Question 15

Not yet answered

Marked out of 2.00

Flag question

X	-10	-20	30
P(X)	1/5	3/10	1/2

Let X be a random variable with the probability distribution given above. The mean of $g(X) = 2X$ is:

Select one:

- 1
- 7
- 14
- -1
- None of the above

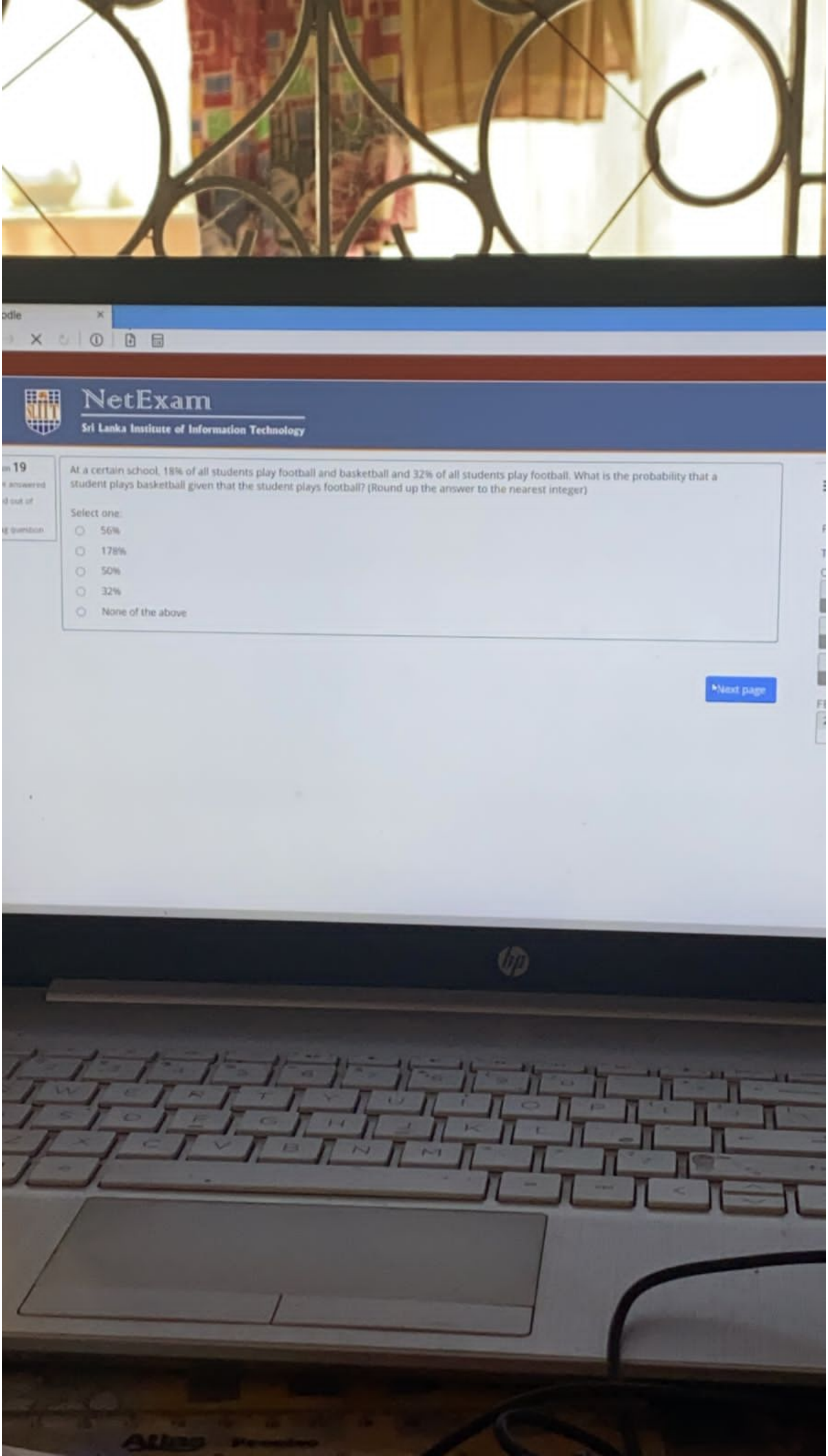
The below transformation is applied to a data set

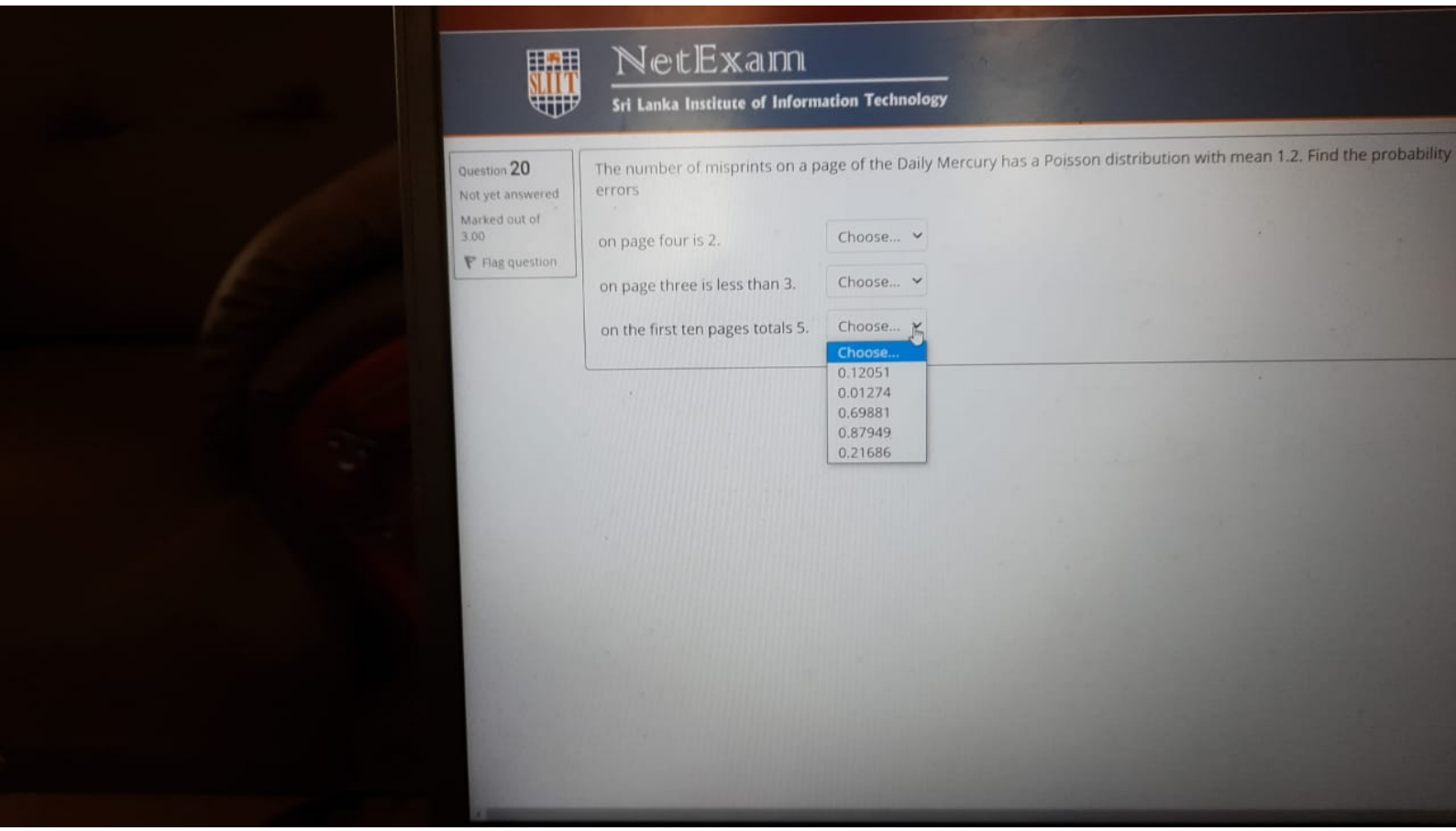
$Y = 3X + 4$

Where X is old data and Y is New data. If the variance of the old data set is 5. The variance of the new data set is,

Select one:

- 49
- 5
- 9
- 45
- None of the above


NetExam
Sri Lanka Institute of Information Technology
At a certain school, 18% of all students play football and basketball and 32% of all students play football. What is the probability that a student plays basketball given that the student plays football? (Round up the answer to the nearest integer)
Select one:
56%
178%
50%
32%
None of the above
Next page


NetExam
Sri Lanka Institute of Information Technology
Question 20
Not yet answered
Marked out of 3.00
Flag question
The number of misprints on a page of the Daily Mercury has a Poisson distribution with mean 1.2. Find the probability
errors
on page four is 2.
Choose...
on page three is less than 3.
Choose...
on the first ten pages totals 5.
Choose...
Choose...
0.12051
0.01274
0.69881
0.87949
0.21686

The purpose of stratified random sampling is to make certain that,

Select one:

- Every member of the population has an equal chance of being selected for the sample.
- The sample proportionately represents individuals from different categories of the population.
- The participants chosen for the study are the ones most likely to react to the treatment.
- The sample is more representative of the actual population than the accessible population.
- None of the above

NetExam

Sri Lanka Institute of Information Technology

The statistics course consists of two (2) online quizzes. 22% of the class passed both tests and 41% of the class passed the first test. About what percent of those who passed the first test also passed the second test? (Round up the answer to the nearest integer)

Select one:

- 54%
- 20%
- 3%
- 24%
- None of the above

Next pag

NetExamn

Sri Lanka Institute of Information Technology

A personal computer user survey was conducted. Time of personal computer use per week is an example of a

Select one:

- Discrete numerical variable
- Continuous numerical variable
- Nominal categorical variable
- Ordinal categorical variable
- None of the above.

NetExam

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16

nswered

out of

question

A and B are two events. P(A and B)' is equal to;

Select one:

- P(A' or B)
- P(A and B')
- P(A' and B')
- P(A' or B')
- None of the above.

Determine the value of k so that the function P(X=x)= kx for x=1,2,3,4,5,6 can serve as a probability distribution of the dis
variable X.

Select one:

- 3/21
- 5/21
- 1/21
- 2/21
- None of the above

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To be an outlier for the following data set, data points should lie between,

10.2, 14.1, 14.4. 14.4, 14.4, 14.5, 14.5, 14.6, 14.7, 14.7, 14.7, 14.9, 15.1, 15.9, 16.4

Hint: Outlier is defined as a value x, holding the following conditions.

x > Q3 + 1.5*(Inter Quartile Range)

x < Q1 - 1.5*(Inter Quartile Range)

Upper bound Choose...

Lower bound

- Choose...
- 15.35
- 15.25
- 13.15
- 16.15
- 13.65
- 15.65
- 12.45
- 13.45
- 15.45

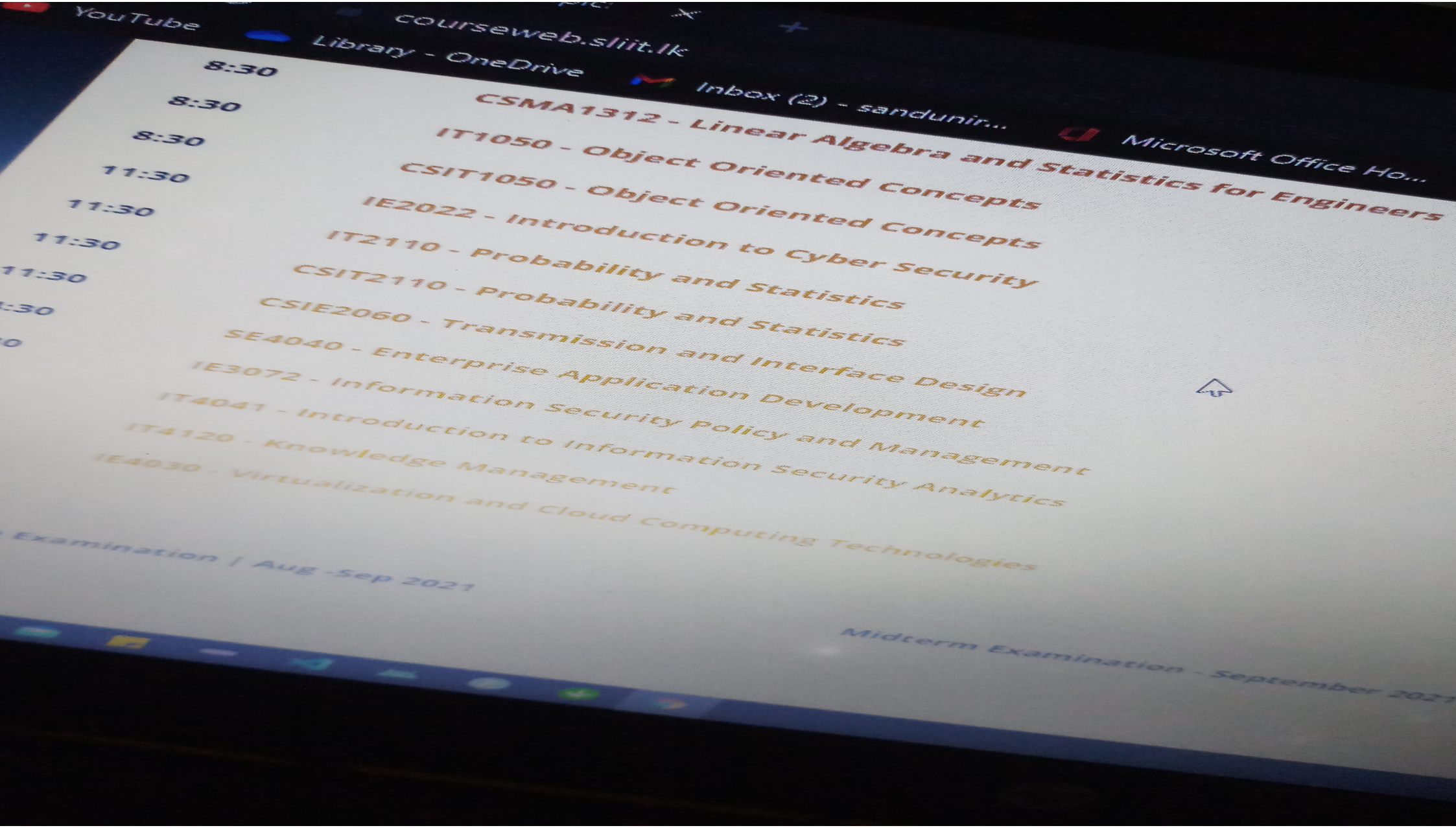
NetExamn

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Let E be an event and E' is its complement. If $P(E) = 1/3$, what is $P(E')$?

Select one:

- $P(E') = P(E) = 1/3$
- $P(E') = P(E) - 1 = -2/3$
- $P(E') = 2 * P(E) = 2/3$
- $P(E') = 1 - P(E) = 2/3$
- None of the above


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IE3072 - Information Security Policy and Management
IT4041 - Introduction to Information Security Analytics
IT4120 - Knowledge Management
IE4030 - Virtualization and Cloud Computing Technologies
Examination | Aug -Sep 2021
Midterm Examination - September 2021

x > Q3 + 1.5*(Inter Quartile Range)

x < Q1 - 1.5*(Inter Quartile Range)

Is "145" an aoutlier? Choose...

Is "18" an outlier

Choose...
No
Yes

After studying a couple's family history, a doctor determines that the probability of any child born to this couple having a gene for disease X is 1 out of 4. If the couple has three children, what is the probability that exactly two of the children have the gene for disease X?

Select one:

- 9/64
- 10/64
- 12/64
- 7/64
- None of the above

Question 7

Not yet answered

Marked out of 1.00

Flag question

This stem and leaf plot shows the number of cookies that Tia's Girl Scout troop sold each week. How many weeks did they sell cookies?

Stem	Leaf
5	1 1 4
6	4 5 6
7	2 2
8	3 6 7 7

Select one:

- 4
- 87
- 12
- 51
- The Stem and leaf plot do not say.

Next page

NetExam

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When each member of a population has an equally likely chance of being selected, this is called:

Select one:

- A nonrandom sampling method.
- A quota sample.
- A judgement sampling
- A simple random sample.
- None of the above.

The type of sampling in which each member of the population selected for the sample is returned to the population before the next member is selected is called,

Select one:

- Sampling without replacement
- Sampling with replacement
- Simple random sampling
- Systematic sampling
- None of the above

Next page